The Clebsch cubic
Recovering, orienting, and realizing — **II**

Quadratic trade rigidity and cubic orientation in conic matching quotients

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From deep holes, the cubic takes shape, ***finds its bearings****, and stands fixed while its shadows move.*

Abstract

Restriction to a conic forgets how its marked points were paired into secants. We classify the full $\mathrm{PGL}_2(q)$-orbits for which the conic-quotient evaluation space has a one-dimensional strength-two trade space with two-valued generator: they are precisely the balanced $B_3/\mathbb{F}_7$ and $H_3/\mathbb{F}_{11}$ orbits. The proof derives the $q+q$ split from modular sheet structure; a Frobenius-digit criterion and one first-wall spill replace extension-field casework. For the two surviving configurations, quadratic products recover the unordered factorization sheets. Their radial nonvanishing has one geometric source: an edge selects two cross-sheet matchings whose complements form a single alternating cycle, and a Dickson recurrence handles both cycle lengths. The first nonzero signed tensor moment is an anti-invariant cubic line. The resulting 14- and 22-point configurations are self-associated and arithmetically Gorenstein; maximal isotropy identifies the perfect Artinian pairing, and the signed cubic is its inverse system. The $A_3/\mathbb{F}_5$, $B_3/\mathbb{F}_7$, and $H_3/\mathbb{F}_{11}$ quotient spans have dimensions $3, 6, 10$; in type H_3, the only missing Fischer layer is $Q\mathcal{H}_2$.

Keywords. conic ideal; secant matching; Coxeter configuration; self-associated points; arithmetically Gorenstein; cubic inverse system; modular representation.

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1 Introduction

We begin with marked secant geometry and ask a complementary reconstruction question: what does common restriction to a conic forget, and what does the conic-ideal quotient remember?

The answer is governed by a short chain of mechanisms. A four-endpoint switch shows that competing secant products differ by the conic equation, so their quotients form an affine configuration. In the balanced cases, Hadamard squares recover an unordered bipartition of that configuration; the first signed tensor that survives is cubic and orients the two parts. Thus quadratic data recover the factorization without choosing its orientation, and the first odd covariant restores that choice.

type	linear quotient	sheets/cubic
$A_3/\mathbb{F}_5$	rank 3	not applicable
$B_3/\mathbb{F}_7$	rank 6	recovered/oriented
$H_3/\mathbb{F}_{11}$	rank 10	recovered/oriented

The proof has three layers. Section 2 proves the quotient and switch identities symbolically for arbitrary matchings. Section 3 first classifies every full projective matching orbit with a two-valued one-dimensional quadratic trade. The classification combines the projective sheet module with a uniform first-wall obstruction; only a local calculation in four restricted modules survives from the extension-field problem. The same section then obtains all three ranks from the affine connecting cocycle and the defining-characteristic SL_2-modules in the Fischer decomposition. One edge-selected alternating-cycle calculation supplies the radial line in both B_3 and H_3. In Section 4, the defining-characteristic permutation module proves the sheet restriction and moment ranks, and elementary duality forces the nonzero cubic and full Schur cube. Signed Gale duality records self-association, while maximal-isotropic quotient duality gives the Gorenstein pairing directly. Appendix F maps each finite input to its primary check and independent replay.

Theorem 1.1 (Quadratic recovery and cubic orientation)**.** *Let the A_3, B_3, and H_3 matching configurations be the finite matching orbits defined in Section 3. Then:*

(i) *matching products scaled as in Section 2 have equal restriction to the conic, and their conic-ideal quotients have difference ranks* 3, 6, *and* 10*;*

(ii) *among all odd prime powers q, the $B_3/\mathbb{F}_7$ and $H_3/\mathbb{F}_{11}$ configurations are the only full* $\mathrm{PGL}_2(q)$*-orbits whose strength-two trade space is one-dimensional and generated by a vector with exactly two values;*

(iii) *in the B_3 and H_3 cases, the two factorization sheets are, up to interchange, the unique complementary halves with equal first and second tensor moments;*

(iv) *in those two cases, the signed moments vanish through degree two, while a nonzero degree-three tensor transforms by the character that exchanges the sheets;*

(v) *the homogenized B_3 and H_3 configurations are self-associated with Gorenstein coordinate rings; the signed cubic is the Macaulay inverse system of their Artinian reductions.*

Appendix B gives a modular interpretation of the H_3 affine-origin obstruction and its homogeneous extension. Sections 2–4 prove the quotient, rank, and quadratic/cubic recovery results; Appendices A–E record H_3 refinements, and Appendix F documents verification.

2 Conic matching products

Write the standard conic as

$$\mathcal{C} = V(Q), \qquad Q = XZ - Y^2,$$

and parametrize it by $\nu(s:t) = (s^2 : st : t^2)$. For points $a = (a_0 : a_1)$ and $b = (b_0 : b_1)$ of $\mathbb{P}^1$, put $[a, b] = a_0 b_1 - a_1 b_0$. We scale the secant L_{ab} through $\nu(a)$ and $\nu(b)$ by the condition

$$L_{ab}(\nu(s:t)) = [a, (s:t)]\,[b, (s:t)].$$

Let Ω be a set of $2m$ distinct marked points of $\mathbb{P}^1$, with fixed vector representatives. For a perfect matching M of Ω, define its matching product

$$P_M = \prod_{\{a,b\}\in M} L_{ab}.$$

The vector representatives are part of the marked-conic data. Their role is to make the products comparable before passing to projective equivalence.

Proposition 2.1 (The matching-secant quotient)**.** *For every $m \geq 2$ and every two perfect matchings M, N of Ω,*

$$P_M|_{\mathcal{C}} = P_N|_{\mathcal{C}}, \qquad P_M - P_N \in (Q).$$

Consequently, after choosing a reference matching M_0,

$$\Phi_{M_0}(M) = \frac{P_M - P_{M_0}}{Q} \in H^0(\mathbb{P}^2, \mathcal{O}(m-2)) \tag{2.1}$$

is an affine configuration. Changing the reference matching translates this configuration. If $g \in \mathrm{GL}_2$ preserves Ω, $\rho(g)$ is its action on binary quadratics, and representatives are transported by g, then

$$\Phi_{gM_0}(gM) \circ \rho(g) = \det(g)^{2m-2} \Phi_{M_0}(M). \tag{2.2}$$

Thus the difference space, its rank, and its affine moment conditions do not depend on M_0 and are equivariant under the endpoint stabilizer.

Proof. For $x \in \mathbb{P}^1$,

$$P_M(\nu(x)) = \prod_{a \in \Omega} [a, x],$$

which does not depend on the pairing. The kernel of the Veronese pullback

$$k[X, Y, Z] \longrightarrow k[s, t], \qquad (X, Y, Z) \longmapsto (s^2, st, t^2)$$

is the principal ideal (Q), so Q divides every difference. Changing the reference follows by subtraction. Finally, $L_{ga,gb} \circ \rho(g) = \det(g)^2 L_{ab}$ and $Q \circ \rho(g) = \det(g)^2 Q$, which gives (2.2). ☐

2.1 The four-endpoint switch

For four distinct endpoints, the Plücker relation gives the local switch identity

$$L_{ab} L_{cd} - L_{ac} L_{bd} = [a, d][b, c]\, Q. \tag{2.3}$$

Multiplying (2.3) by the secant factors common to two matchings proves divisibility for a single switch. Any two perfect matchings are joined by a sequence of such switches: if M and N differ at a, write $\{a, b\} \in M$ and $\{a, c\} \in N$; switching the two M-edges through b and c creates $\{a, c\}$ and reduces the symmetric difference. This also gives an elementary proof of the divisibility clause in Proposition 2.1.

The switch calculation and the global pullback proof play different roles. The pullback proves the quotient for arbitrary matchings at once. The switch identity records how local changes of factorization move in the quotient space. A selected Coxeter orbit need not contain every perfect matching or every intermediate switch.

3 The rank-three configurations

We now select three finite matching orbits. This selection is discrete input: Proposition 2.1 applies to every matching, but it does not distinguish the Coxeter orbits. Here and below, "rank three" refers to Coxeter rank. It does not refer to the dimensions of the conic-ideal quotient spaces, which are $3, 6, 10$.

For $q \in \{5, 7, 11\}$, index

$$\mathbb{P}^1(\mathbb{F}_q) = \{0, 1, \ldots, q-1, \infty\},$$

where a denotes $(1 : a)$ and ∞ denotes $(0 : 1)$. Let $G_q = \mathrm{PGL}_2(q)$ act in the usual way. The three base matchings are

$$\begin{array}{lll} T & q & M_T \\ \hline A_3 & 5 & \{0,\infty\},\{1,4\},\{2,3\}, \\ B_3 & 7 & \{0,2\},\{1,4\},\{3,\infty\},\{5,6\}, \\ H_3 & 11 & \{0,1\},\{2,5\},\{3,7\},\{4,9\},\{6,8\},\{10,\infty\}. \end{array} \tag{3.1}$$

Define the *rank-three matching configuration*

$$\Omega_T = G_q \cdot M_T.$$

The stabilizers of the three displayed matchings are respectively S_4, S_4, A_5, and hence

$$|\Omega_{A_3}| = 5, \qquad |\Omega_{B_3}| = 14, \qquad |\Omega_{H_3}| = 22. \tag{3.2}$$

These marker spaces and their stabilizers are classical. In particular, Edge and Dye identify the exceptional finite conic configurations and substantial parts of their incidence geometry [2, 3]. We use the Coxeter names for the corresponding tetrahedral, octahedral, and icosahedral data; neither the orbit counts in (3.2) nor the ambiguity of an unmarked conic is asserted here as new.

Lemma 3.1 (Projective–trade reduction)**.** *Let k be an algebraically closed field of odd characteristic, let $H \lhd G$ have index two, and let a transitive G-set Ω split as two transitive H-sets $\Omega_+ \sqcup \Omega_-$. Let*

$$0 \longrightarrow F \longrightarrow E \xrightarrow{\epsilon} k \longrightarrow 0$$

be an exact sequence of kG-modules, and let $v : \Omega \to E$ be a G-map with $\epsilon(v_\omega) = 1$. Define

$$\mu : k[\Omega] \longrightarrow \mathrm{Sym}^2 E, \qquad [\omega] \longmapsto v_\omega^2.$$

Also put

$$a : k[\Omega] \longrightarrow E, \qquad [\omega] \longmapsto v_\omega.$$

All embeddings, pullbacks, and splittings below are in the category of kG-modules. Assume that

$$\ker \mu = k \left(\sum_{\omega \in \Omega_+} [\omega] - \sum_{\omega \in \Omega_-} [\omega] \right),$$

and that $k[\Omega_+]$ contains a nontrivial simple H-submodule S such that ${}^gS \simeq S$ for $g \in G \setminus H$, and that S extends to G. If S^+ and $S^- = S^+ \otimes \chi$ are its two G-extensions, where χ is the nontrivial character of G/H, then both S^+ and S^- embed in $\mathrm{Sym}^2 E$. For either summand $T = S^\pm$, let $j = \mu|_T : T \to \mathrm{Sym}^2 E$ be the resulting embedding and put $i = \partial j$. Exactly one of the following occurs:

1. *$i = 0$ and $j(T) \subseteq \mathrm{Sym}^2 F$;*

2. *$i : T \to E$ is an embedding and the pullback along i of*

$$0 \longrightarrow \mathrm{Sym}^2 F \longrightarrow \mathrm{Sym}^2 E \xrightarrow{\partial} E \longrightarrow 0$$

splits, where

$$\partial(xy) = \epsilon(x)y + \epsilon(y)x.$$

Moreover, let $\xi \in \mathrm{Ext}^1_G(k, F)$ *be the class of the affine extension and let* $\pi : F \otimes F \to \mathrm{Sym}^2 F$ *be the canonical quotient,* $\pi(f \otimes f') = ff'$*. Since* T *is nontrivial,* $i(T) \subseteq F$*, and*

$$\delta_T(i) = i^* \pi_*(F \otimes \xi) \quad \textit{in} \quad \mathrm{Ext}^1_G(T, \mathrm{Sym}^2 F). \tag{3.2a}$$

Proof. Choose $g \in G \setminus H$. The G-submodule of $k[\Omega]$ generated by S is $S \oplus {}^g S$. It has zero intersection with the displayed kernel, since that kernel is trivial as an H-module and S is not. Thus μ embeds $S \oplus {}^g S$.

Choose the extension S^+. The elementary tensor identity for induction gives

$$\mathrm{Ind}_H^G S \xrightarrow{\sim} S \oplus {}^g S \simeq S^+ \otimes k[G/H].$$

The first map sends $x \otimes s$ to xs; it is an isomorphism because its two sheet-supported images are disjoint. Since $2 \neq 0$ in k, one has $k[G/H] \simeq k \oplus k_\chi$. Hence the last module is $S^+ \oplus S^-$, proving that both parities embed.

It remains only to inspect the canonical map ∂. It is G-equivariant, and on every basis vector

$$\partial\mu([\omega]) = 2v_\omega = 2a([\omega]).$$

Choose $e \in E$ with $\epsilon(e) = 1$, merely as a vector-space splitting. Then

$$\mathrm{Sym}^2 E = ke^2 \oplus eF \oplus \mathrm{Sym}^2 F, \qquad \partial(e^2) = 2e, \quad \partial(ef) = f.$$

Consequently the displayed sequence is exact. Since T is simple, i is either zero or injective. In the first case the image of j lies in $\mathrm{Sym}^2 F$. In the second, the map $s \mapsto (j(s), s)$ is a section of the pullback of that sequence along i. Equivalently, if

$$\delta_T : \mathrm{Hom}_G(T, E) \longrightarrow \mathrm{Ext}^1_G(T, \mathrm{Sym}^2 F)$$

is the connecting map, then $\delta_T(i) = 0$. This proves the dichotomy. We use the convention that $\delta_T(i)$ is the Yoneda class of the pullback along i.

For the final assertion, put $D = \partial^{-1}(F)$. Multiplication $f \otimes x \mapsto fx$ gives a morphism from

$$0 \longrightarrow F \otimes F \longrightarrow F \otimes E \xrightarrow{1 \otimes \epsilon} F \longrightarrow 0$$

onto

$$0 \longrightarrow \mathrm{Sym}^2 F \longrightarrow D \xrightarrow{\partial} F \longrightarrow 0.$$

The left map is π, the right map is the identity, and the lower row is the pushout of the upper row: this is immediate from the vector-space splitting $E = ke \oplus F$ used above. Thus the lower row represents $\pi_*(F \otimes \xi)$. Since $\epsilon i : T \to k$ is zero for nontrivial simple T, the map i lands in F; pulling back the lower row gives (3.2a). More concretely, choose the same vector-space lift e and put $c(g) = ge - e \in F$. With the connecting-cocycle convention $g \mapsto gsg^{-1} - s$, the class in (3.2a) is represented by

$$z_i(g) : T \longrightarrow \mathrm{Sym}^2 F, \qquad z_i(g)(t) = i(t)c(g). \tag{3.2b}$$

Changing e changes z_i by a coboundary. $\square$

The all-field exclusion now has three steps. A p'-stabilizer supplies an actual simple socle of the self-dual sheet module. The two outer extensions of that socle are forced into the quadratic moment module, but one parity is absent from $\mathrm{Sym}^2 F$. If its image under ∂ vanishes, this is already a contradiction; otherwise the affine cocycle contracts either to its original class or to the unique two-coordinate first-wall obstruction. The next lemma proves the last two claims before we return to matching stabilizers.

Lemma 3.2 (Lucas socle, square parity, and the first wall). *Let $q = p^e$ be odd, let $H = \mathrm{PSL}_2(q)$, and put*

$$d = \frac{q-3}{2}, \qquad a = \frac{p-3}{2}, \qquad b = \frac{p-1}{2}.$$

Over $k = \overline{\mathbb{F}}_q$, write $L(c_0, \dots, c_{e-1})$ for $L(c_0) \otimes L(c_1)^{(1)} \otimes \cdots$, and put $W = \nabla(d)$. Under modular Hermite reciprocity,

$$F = \mathrm{Sym}^d L(2) \simeq \mathrm{Sym}^2 W.$$

The following statements hold.

1. *A simple $L(c_0, \dots, c_{e-1})$ occurs in the socle of F if and only if, for every j, there are $r_j \geq 0$ and $\delta_j \in \{0, 1\}$ such that*

$$c_j = 2n_j - 2\delta_j - 4r_j, \qquad n_0 = a, \quad n_j = b \ (j > 0),$$

   and $\sum_j \delta_j$ is even. These data give a basis of the actual finite-group Hom space, not only a list of composition factors.

2. *Let $S = L(c_0, \dots, c_{e-1})$ be any q-restricted simple which descends to H. It is invariant under the diagonal outer automorphism and has two extensions to $G = \mathrm{PGL}_2(q)$. There is a distinguished extension $S^\square$, opposite to the determinant-normalized divided-power parity, for which*

$$\mathrm{Hom}_G(S^\square, \mathrm{Sym}^2 F) = 0,$$

   and the other extension is $S^\square \otimes \chi$, where χ is the nontrivial character of G/H.

3. *Let $S = L(s)$, where s is even and $0 \leq s < p - 1$. Then*

$$\dim \mathrm{Hom}_H(S, F) = \begin{cases} 1, & eb \equiv 1 + s/2 \pmod 2, \\ 0, & eb \not\equiv 1 + s/2 \pmod 2. \end{cases}$$

   If the Hom line has the parity $S^\square$, then the pullback along its inclusion $i : S^\square \hookrightarrow F$ of

$$0 \longrightarrow \mathrm{Sym}^2 F \longrightarrow \mathrm{Sym}^2 E \xrightarrow{\partial} E \longrightarrow 0$$

   is nonsplit. More generally, if $i : S \hookrightarrow F$ has a retraction $\rho : F \to S$, the class contracts to

$$(\dim S)[c] + i_* \rho_* [c] \in H^1(H, F).$$

   In particular the pullback is nonsplit whenever this class is nonzero.

Proof. Steinberg's tensor-product theorem gives the notation and the complete list of q-restricted simples [7, Theorems 1.1 and 7.4]. McDowell and Wildon's modular Hermite reciprocity gives the displayed isomorphism [5, Corollary 1.5]; here the second symmetric power is unambiguous because p is odd.

We record the finite-group seam in the coefficient argument. In the divided-power basis, the equation saying that a highest vector is fixed by the positive root group is a polynomial in its parameter of degree at most $2d = q - 3$. It vanishes on $\mathbb{F}_q$ precisely when all its coefficients vanish. Lucas's theorem separates those equations into the e base-p digits of

$$d = a + bp + \cdots + bp^{e-1}.$$

In one digit, the symmetric and alternating Clebsch–Gordan lines have highest weights $2n-4r$ and $2n-2-4r$, respectively, and the tensor flip acts on the corresponding line by $(-1)^{\delta_j}$. The product lies in $\mathrm{Sym}^2 W$ exactly when the product of these signs is $+1$. Conversely, tensoring the one-digit divided-power maps produces every displayed embedding. Thus the argument solves the root-group intertwining equations themselves and proves the first assertion about Hom_H.

The same calculation with two copies of $F \subset W \otimes W$ gives the outer assertion. Normalize $L(c_0, \dots, c_{e-1})$ as the $\mathrm{PGL}_2(q)$-module

$$\bigotimes_j (\mathrm{Sym}^{c_j} k^2)^{(j)} \otimes \det^{-\frac{1}{2}\sum_j p^j c_j}.$$

Conjugation by a nonsquare diagonal matrix acts on a Lucas product by the product of its local flip signs. In $\mathrm{Sym}^2(\mathrm{Sym}^2 W)$ the two inner flips and the outer flip are all $+1$. Hence every H-map to $\mathrm{Sym}^2 F$ has one determinant-normalized parity, whereas twisting its source by χ negates the conjugation equation. We denote this opposite, absent extension by $S^\square$. There is one extra point because the root polynomial for four tensor factors can cross q. Its degree is below $2q$, so division by $t^q - t$ leaves only the first wall. Replacing a root exponent q by 1 changes the determinant-normalizing exponent by $(q-1)/2$; for a nonsquare ν, Euler's criterion gives $\nu^{(q-1)/2} = -1$. The corresponding dual-Weyl crossing reverses exactly one local tensor-flip sign. The two sign changes cancel. Thus the first-wall rows have the same outer parity as the ordinary Lucas rows. This proves the asserted G-Hom vanishing for the full finite-group Hom space, not merely for its algebraic-group part.

For $S = L(s)$, the local signs are unique:

$$\delta_0 \equiv a - s/2, \qquad \delta_j \equiv b \quad (j > 0) \pmod 2.$$

Their sum is even exactly under the displayed congruence, and uniqueness also proves that the Hom space is a line.

For the contraction assertion, let

$$D_\rho : \mathrm{Sym}^2 F \longrightarrow S \otimes F, \qquad D_\rho(xy) = \rho(x) \otimes y + \rho(y) \otimes x.$$

Compose D_ρ with a cochain $S \to \mathrm{Sym}^2 F$, then take the categorical trace over S. This is an H-map

$$C_{\rho,i} : \mathrm{Hom}(S, \mathrm{Sym}^2 F) \longrightarrow F.$$

On $z_i(g)(t) = i(t)c(g)$, the first summand of D_ρ closes the S-loop and the second projects $c(g)$. Hence

$$C_{\rho,i}(z_i)(g) = (\dim S)c(g) + i\rho(c(g)).$$

The formula commutes with coboundaries and proves the asserted identity in H^1. This is the direct evaluation–coevaluation contraction; it uses a retraction of the selected copy inside F, not a retraction $E \to S$.

It remains to handle a nonretracted occurrence. Let $B = U \rtimes T_0$ be the split Borel of H. Since $[H : B] = q + 1$ is invertible in k, restriction on H^1 is injective. Normalize the affine cocycle $c(g) = ge - e$ to vanish on T_0.

Assume now $e > 1$ and put

$$T_1 = L(p-2, 1), \qquad R = L(p-2-s, 1), \qquad Y = L(0, 2), \qquad r = p - 2 - s.$$

The first dual-Weyl wall of W, followed by the two symmetric-square filtrations above, gives a canonical two-coordinate quotient of the Borel coboundary equation for $z_i(g)(t) = i(t)c(g)$. We describe it explicitly, so no filtration claim is hidden; the underlying dual-Weyl short exact sequence is the one recorded in [6, Lemma 2.3]. The affine class has one first-wall component, T_1. Indeed the torus-normalized root cocycle has no ordinary Lucas component; its sole term divisible by $t^q - t$ has digit weights $(p-2, 1)$ and coefficient 1. Equivalently, $H^1(B, T_1)$ is a line and every other factor at this first wall has zero T_0-fixed U-cohomology. The only cochain row which can meet it is the map $S \to R \otimes T_1$ whose zeroth-digit part is

$$v_m \longmapsto \sum_{j=0}^{r} (-1)^j \binom{r}{j} v_j \otimes w_{r-j+m},$$

and whose first-digit part is the bracket $L(0) \to L(1) \otimes L(1)$. The formula follows directly by applying the positive-root operator: consecutive coefficients satisfy $c_{j+1} = -(r-j)c_j/(j+1)$. Thus its seed is $(1-z)^r$. The row is forced: in the zeroth digit the Clebsch–Gordan equation

$$s = r + (p-2) - 2k$$

gives $k = r$, and in the first digit the displayed bracket is unique. Conversely, solving the two digit equations for a row reaching $Y \otimes R$ recovers these same values. Hence no other first-wall row can meet the spill coordinate. The first-wall trace and the adjacent spill are respectively

$$-[z](1-z)^r = r, \qquad [1](1-z)^r = 1.$$

All later Frobenius digits have their unique Lucas signs and tensor with this row; they create no further row meeting either coordinate. Therefore, if a Borel cochain killed z_i, the coefficient x of this row would have to satisfy simultaneously

$$rx = 1, \qquad x = 0.$$

The second equation is the $Y \otimes R$ component. No other row reaches it, and it admits no Borel cochain from S. Indeed the weights of $Y \otimes R$ are

$$p(2-2j) + (p-2-s-2k) + p(1-2\ell),$$

with $0 \le j \le 2$, $0 \le k \le p-2-s$, and $0 \le \ell \le 1$. The coefficient-1 weights begin at $s+2$, the coefficient-(-1) weights end at $-s-2$, and the coefficient-± 3 weights are farther away. Their differences from the weights of S have absolute value below $q-1$ in every case used here, so $\mathrm{Hom}_B(S, Y \otimes R) = 0$. Finally $1 \le r < p$, so the two equations are incompatible. This two-coordinate contraction proves that the restriction of the pullback class is nonzero, and Borel injectivity proves the assertion over H, hence over G. □

Lemma 3.3 (Uniform nonprincipal-sheet exclusion)**.** *Let $q = p^e$ be odd, let $G = \mathrm{PGL}_2(q)$ and $G^+ = \mathrm{PSL}_2(q)$, and let Ω be a full G-orbit of perfect matchings split into two G^+-orbits. Let $L \subseteq \mathbb{F}_q^\Omega$ be its affine evaluation space. If $(L^{\circ 2})^\perp \subseteq \mathbb{F}_q^\Omega$ is the sheet-sign line, then either sheet has q members.*

Proof. We give the modular argument because it removes the one-factorization hypothesis. Let $K \le G^+$ stabilize a matching and write a sheet as G^+/K. A nontrivial unipotent cannot fix a perfect matching: it has one fixed endpoint, while the partner of that endpoint would have an orbit of length p. Thus K is a p'-group and

$$P_0 = \mathbb{F}_q[G^+/K]$$

is projective. Its restriction to a translation Sylow subgroup is a sum of regular modules, and

$$|G^+/K| = q\lambda \tag{3.2c}$$

for some positive integer λ. Extend scalars to $k = \overline{\mathbb{F}}_q$ and put $P = k \otimes_{\mathbb{F}_q} P_0$.

Let L_k be the scalar extension of the affine evaluation space and put $E = L_k^*$. The inclusion of the constants in L_k dualizes to the point-vector extension

$$0 \longrightarrow F \longrightarrow E \xrightarrow{\epsilon} k \longrightarrow 0, \qquad F = \mathrm{Sym}^{(q-3)/2} L(2), \tag{3.2d}$$

and evaluation at a matching M is a vector $v_M \in E$ with $\epsilon(v_M) = 1$. The transpose of the multiplication–evaluation map $\mathrm{Sym}^2 L_k \to k^\Omega$ is the moment map in Lemma 3.1. Thus its kernel is the scalar extension of the strength-two trade space; here we use that scalar extension is flat and commutes with Sym^2. Write μ for this moment map and $a : k[\Omega] \to E$ for the linear moment map $[M] \mapsto v_M$.

We apply Lemma 3.2. Its absent outer extension will be denoted $S^\square$. Write

$$j_\square = \mu|_{S^\square}, \qquad i_\square = \partial j_\square = 2a|_{S^\square}.$$

If $i_\square = 0$, then $j_\square$ would embed $S^\square$ in $\mathrm{Sym}^2 F$, contrary to the square-parity vanishing. If $i_\square \neq 0$ and $S = L(s)$ is one-digit, then $j_\square$ would split its quadratic pullback, contrary to the first-wall conclusion. Thus it remains to produce the stated simple submodule for every stabilizer with $\lambda > 1$, and to check that every multi-digit choice below is absent from the affine socle.

Dickson's p'-subgroup list reduces K to cyclic, dihedral, and exceptional types; compare the maximal-subgroup form in [4, Theorem 2.2]. For a cyclic torus subgroup take $S = L(2)$. If the Weyl involution is present, take $S = L(4)$ for $p \geq 7$, and take $S = L(2,2)$ when $p = 3, 5$ and $e > 1$. The zero-weight vector is fixed in each case: its Weyl sign is -1 in $L(2)$, $+1$ in $L(4)$, and $+1$ in the two-digit tensor product. In characteristic three the zeroth digit of the affine socle is 0, so neither $L(2)$ nor $L(2,2)$ can occur in F. Thus $i_\square = 0$, including at $q = 9$. This is the structural characteristic-three endpoint.

For A_4, S_4, A_5, averaging the Steinberg-digit character over the binary-polyhedral lift gives, respectively,

range	S
$A_4,\ p \geq 11$	$L(6)$
$S_4,\ p \geq 11$	$L(8)$
$A_5,\ p \geq 17$	$L(12)$
$A_4,\ p = 5, 7,\ e > 1$	$L(1,1)$
$S_4,\ p = 5,\ e > 1$	$L(2,2)$
$S_4,\ p = 7,\ e > 1$	$L(1,1)$
$A_5,\ p = 7, 11,\ e > 1$	$L(3,3)$, $L(1,1)$ respectively
$A_5,\ p = 13,\ e > 1$	$L(1,7)$.

(3.2e)

In every row $\dim S^K = 1$. This follows directly from

$$\dim S^K = \frac{1}{|K|} \sum_C |C| \prod_j \sum_{\ell=0}^{c_j} \zeta_C^{p^j(c_j - 2\ell)},$$

where ζ_C is the Teichmuller lift of an eigenvalue of a binary-polyhedral representative. Use lift orders $1, 4, 6$ for A_4, $1, 4, 6, 8$ for S_4, and $1, 4, 5, 6, 10$ for A_5. More precisely, the (class size, lift order) pairs are

$$\begin{array}{c|l} A_4 & (1,1), (3,4), (8,6) \\ S_4 & (1,1), (9,4), (8,6), (6,8) \\ A_5 & (1,1), (15,4), (20,6), (12,10), (12,5). \end{array}$$

Thus the assertion concerns the simple module itself.

Frobenius reciprocity makes S a quotient of $P = k[G^+/K]$. The permutation form identifies P with P^*, and every displayed S is self-dual, so dualizing embeds S in P. This proves simple-submodule existence even if the principal projective cover is a summand of P. The digit sum is even, and

$$\bigotimes_j (\mathrm{Sym}^{c_j} k^2)^{(j)} \otimes \det^{-\frac{1}{2}\sum_j p^j c_j}$$

is an extension to G; the other is its χ-twist.

For $S = L(s)$, with $s = 2, 4, 6, 8$, or 12, the Lucas criterion and the first-wall contraction in Lemma 3.2 give the required obstruction. In a prime field the Fischer summands below the unique affine top are direct:

$$F = \bigoplus_{0 \le j \le \lfloor (p-3)/4 \rfloor} L(p - 3 - 4j).$$

Indeed $d < p$, so the symmetric-power idempotent makes F a tilting summand of $L(2)^{\otimes d}$; every displayed highest weight is in the bottom alcove, where its tilting module is simple. The affine class is supported only on the top factor. The contraction therefore gives $(\dim S)[c]$ on a lower factor and $(\dim S + 1)[c]$ on the top factor. Both scalars are nonzero in every selected row. To see the support assertion directly, restrict to the prime-field Borel. On $L(m)$, $U = C_p$ is one Jordan block, so $H^1(U, L(m))$ is its coinvariant line; the torus acts on that line by $a^{-(m+2)}$. For even $0 \le m \le p - 3$, it is fixed exactly when $m = p - 3$. Restriction on H^1 is injective because the Borel index is $p+1$. Every small-characteristic head in (3.2e) has an odd digit and is absent from the affine socle, except $L(2, 2)$; its unique sign word is alternating, so it too is absent. Hence $i_\square = 0$ in those rows. The $(q, K) = (5, A_4)$ and $(7, A_4)$ endpoint matchings are the unique two-point block systems of the endpoint actions and are preserved by the outer S_4-normalizer; their full orbits therefore do not give a split $\lambda > 1$ case. The characteristic-three torus rows were disposed of above by $L(2)$ and $L(2, 2)$, including $q = 9$; for $q = 3$, the three perfect matchings form one G^+-orbit, so the two-sheet premise is impossible. The isolated $q = 5$ dihedral matching has an outer full stabilizer. We have therefore produced the quadratic pullback obstruction for every possible split stabilizer with $\lambda > 1$, a contradiction. Hence $\lambda = 1$. $\square$

Theorem 3.4 (Uniform balanced-orbit classification)**.** *Let q be an odd prime power, put*

$$G = \mathrm{PGL}_2(q), \qquad G^+ = \mathrm{PSL}_2(q),$$

and let Ω be a G-orbit of perfect matchings of $\mathbb{P}^1(\mathbb{F}_q)$. Form its conic-ideal quotient configuration and let $L \subseteq \mathbb{F}_q^\Omega$ be its affine evaluation space. Put

$$L^{\circ 2} = \mathrm{span}\{fg : f, g \in L\},$$

with pointwise multiplication, and take its annihilator under the standard coordinate pairing on $\mathbb{F}_q^{\Omega}$*. Suppose* $(L^{\circ 2})^{\perp}$ *is one-dimensional, its nonzero vectors have exactly two fibers. Then*

$$(q,\Omega) = (7,\Omega_{B_3}) \quad \textit{or} \quad (q,\Omega) = (11,\Omega_{H_3}). \tag{3.2l}$$

Conversely, both displayed orbits satisfy these intrinsic hypotheses. For either orbit, the two fibers are its two one-factorization sheets.

Proof. Proposition 2.1 makes L, its square, and the line

$$R = (L^{\circ 2})^{\perp}$$

G-stable. Choose $0 \neq \tau \in R$. Its zero set is G-invariant, so transitivity gives τ full support. The two level sets of τ depend only on R, hence G permutes them. This action is nontrivial because G is transitive on Ω.

For $q > 3$, the commutator subgroup of G is G^+, and G/G^+ has order two. The kernel of the action on the two levels is therefore G^+. The stabilizer of a point fixes its level and so lies in G^+; consequently the two levels are exactly the two G^+-orbits. An outer element acts on R by a scalar and exchanges the two nonzero values of τ, so those values are opposites.

Lemma 3.3 shows that each intrinsic level has size q. Thus $|\Omega| = 2q$. When $q = 3$, there are only three perfect matchings altogether, so such an orbit cannot occur.

Now fix $M \in \Omega$ and let $K = \mathrm{Stab}_G(M)$. Orbit–stabilizer and the derived two-sheet conclusion give

$$|K| = \frac{q^2-1}{2}, \qquad K \leq G^+, \qquad [G^+ : K] = q. \tag{3.2m}$$

Assume $q \geq 5$, and place K in a maximal subgroup of G^+. Dickson's maximal-subgroup theorem, in the form recorded by Giudici [4, Theorem 2.2], leaves Borel, dihedral, subfield, and exceptional A_4, S_4, A_5 subgroups. The first two have order smaller than $(q^2-1)/2$. A proper subfield subgroup is no larger than $\mathrm{PGL}_2(r)$, where $q = r^e$ and $e \geq 2$, and

$$r(r^2-1) < \frac{r^{2e}-1}{2}.$$

Thus it too is too small. An exceptional maximal subgroup has order at most 60, so (3.2m) leaves $q = 5, 7, 9, 11$. For $q = 9$, the required order 40 divides none of $12, 24, 60$. In the other three cases, order and divisibility force

$$(q,K) = (5,A_4),\ (7,S_4),\ (11,A_5). \tag{3.2n}$$

It remains to ask which of these subgroup actions preserves a matching. This is a block-system question, not a matching census. For $q = 5$, the A_4-action on the six endpoints is transitive with point stabilizer C_2. Its normal Klein four subgroup gives the unique two-point block system. The normalizer of A_4 in $\mathrm{PGL}_2(5)$ is S_4; uniqueness makes this whole normalizer preserve the matching. Its full orbit therefore has five points, namely Ω_{A_3}, rather than ten. In particular, the other ten matchings cannot form a balanced $5+5$ orbit.

For $q = 7$, the S_4-action on eight endpoints has point stabilizer C_3. A two-point block stabilizer must have order six, and the unique possibility containing that C_3 is its normalizer S_3. Hence the invariant matching is unique. Similarly, for $q = 11$, the A_5-action on twelve endpoints has point stabilizer C_5; its unique possible two-point block stabilizer is $N_{A_5}(C_5) = D_{10}$. The invariant matching is again unique. In each case the two G^+-classes of the exceptional subgroup are fused by G, and the subgroup has no outer normalizer [4]. Its full matching stabilizer is therefore exactly S_4 or A_5, so the orbit has size $2q$ and splits into the two claimed G^+-orbits. The unique block systems are precisely the matchings displayed in (3.1), up to G-conjugacy.

Conversely, $q = 7, 11$ are both 3 (mod 4), so G^+ is transitive on unordered pairs of endpoints. Each q-matching sheet therefore covers every edge equally often; its total of $q(q+1)/2$ edge occurrences equals the number of edges, so it is a one-factorization. Propositions 4.1 and 4.2 identify $(L^{\circ 2})^\perp$ with the sheet-sign line. Hence both displayed orbits satisfy the intrinsic hypotheses. □

Fix M_T as reference and apply (2.1) to every member of Ω_T. Write $R_d = \mathbb{F}_q[X, Y, Z]_d$ and

$$\Delta_Q = 4\partial_X \partial_Z - \partial_Y^2, \qquad \mathcal{H}_d = \ker(\Delta_Q : R_d \longrightarrow R_{d-2}). \tag{3.3}$$

The operator Δ_Q is the Laplacian attached to the dual quadratic form. In the degrees used below, $\mathcal{H}_d$ is the top harmonic summand, while powers of Q give radial summands.

Lemma 3.5 (The edge-selected alternating cycle)**.** *For $q = 7, 11$, let $\Omega_+ \sqcup \Omega_-$ be the two one-factorization sheets. Every endpoint edge e lies in a unique pair $M_+(e) \in \Omega_+$, $M_-(e) \in \Omega_-$. The matchings share only e, and after deleting it their union is one alternating cycle on the other $q - 1$ endpoints. The corresponding conic quotient has nonzero deepest radial projection.*

Proof. Move e to $\{0, \infty\}$, so its secant is Y, and put

$$n = \frac{q-1}{2}, \qquad Q_0 = (\mathbb{F}_q^\times)^2.$$

The complementary matchings have the common torus normal form

$$\{\{u, cu\} : u \in Q_0\}, \qquad \{\{u, c^{-1}u\} : u \in Q_0\}, \tag{3.3a}$$

where $c^2 = 1/4$. Multiplication by $c^{-2} = 4$ has order n on Q_0 for $q = 7, 11$, proving the alternating-cycle assertion.

Write

$$A_c(X, Y, Z) = \prod_{u \in Q_0} (cu^2 X - (1 + c)uY + Z).$$

The two matching products share the factor Y, and equality on the conic gives

$$A_{c^{-1}} - A_c = (XZ - Y^2)\Psi_c. \tag{3.3b}$$

If

$$h_j = \frac{n}{n-j}\binom{n-j}{j},$$

the resultant with $u^n - 1$ gives one Dickson recurrence:

$$A_{c^{-1}} - A_c = 2 \sum_{j=0}^{(n-1)/2} (-1)^j h_j c^j (1 + c)^{n-2j} (XZ)^j Y^{n-2j}. \tag{3.3c}$$

For $q = 7$, division in (3.3b) gives $\Psi_c = 5Y$. For $q = 11$, it gives

$$\Psi_c = 2Y^3 + XZY, \qquad \Delta_Q \Psi_c = 3Y.$$

The full quotient is $Y\Psi_c$, up to the fixed nonzero normalization. The common secant Y has nonzero dual conic norm, so the displayed linear deepest traces force a nonzero radial Fischer component in both degrees. □

Remark 3.6 (The Paley carrier). Label either sheet by its regular translation subgroup. The cross-sheet incidence matrix is the circulant support of $\{0\} \cup (\mathbb{F}_q^\times)^2$, the complement of the Paley difference set. Its bordered sign matrix is the Paley Hadamard matrix of order $q+1$. This explains the multiplier 4 and the sheet orientation, but it is not the Gorenstein pairing used below.

Theorem 3.7 (The $3, 6, 10$ quotient ranks)**.** *For the matching configurations in (3.1), let*

$$W_T = \operatorname{span}_{\mathbb{F}_q}\{\Phi_{M_T}(M) : M \in \Omega_T\}.$$

Then

$$\begin{array}{cccc} T & \deg\Phi & W_T & \dim W_T \\ \hline A_3 & 1 & R_1 = \mathcal{H}_1 & 3, \\ B_3 & 2 & R_2 = \mathcal{H}_2 \oplus \mathbb{F}_7 Q & 6, \\ H_3 & 4 & \mathcal{H}_4 \oplus \mathbb{F}_{11} Q^2 & 10. \end{array} \tag{3.4}$$

If $h_T = 4, 6, 10$ is the Coxeter number and $d = h_T/2 - 1$, the three rows have the uniform form

$$W_T = \begin{cases} \mathcal{H}_d, & d \text{ odd}, \\ \mathcal{H}_d \oplus \mathbb{F}_q Q^{d/2}, & d \text{ even}, \end{cases} \qquad \dim W_T = h_T - 1 + \mathbf{1}_{\{d \text{ even}\}}.$$

In particular, the first two configurations fill the complete quotient form space. The H_3 configuration spans a canonical ten-dimensional subspace of the fifteen-dimensional space R_4.

Proof mode. The quotient construction and all three spanning assertions are proved symbolically. Irreducibility forces the A_3 image and the top B_3 summand. For H_3, a Sylow-normalizer cohomology argument excludes the middle Fischer layer and forces the top harmonic summand. The common edge-selected alternating cycle supplies both radial lines. A separate implementation reconstructs the three full quotient matrices and every edge-selected cycle from the data in (3.1).

Proof. A matching in (3.1) has $m = 3, 4, 6$ edges, respectively, so Proposition 2.1 places its quotient in R_{m-2}, of degrees $1, 2, 4$. Direct differentiation gives

$$\dim \mathcal{H}_1 = 3, \qquad \dim \mathcal{H}_2 = 5, \qquad \dim \mathcal{H}_4 = 9.$$

The radial vector $Q \in R_2$ is not harmonic in characteristic 7.

We first prove the A_3 and B_3 rows without quotient matrices. For $q = 5, 7$, put

$$G = \mathrm{PGL}_2(q), \qquad G^+ = \mathrm{PSL}_2(q), \qquad \chi(g) = \det(g)^{(q-1)/2}.$$

The Dickson form $s^q t - s t^q$, which is the common conic restriction of the matching products, has contragredient weight $\det^{-1}$. The Clebsch identification and Fischer decomposition therefore give

$$\begin{array}{ll} q = 5 & \widetilde{R}_1 = R_1 \otimes \det^{-1} = M_2, \\ q = 7 & \widetilde{R}_2 = R_2 \otimes \det^{-1} = M_4 \oplus M_0, \end{array} \tag{3.4a}$$

where, uniformly,

$$M_r = \operatorname{Sym}^r(V^\vee) \otimes \overset{r/2}{\det} \otimes \chi, \qquad M_0 = \chi.$$

The scalar action is trivial, so these are G-modules. On G^+, M_2 and M_4 are the absolutely irreducible symmetric powers of highest weights $2 < 5$ and $4 < 7$, respectively [10, Fact 3.1].

For either type let

$$c(g) = \Phi_{M_T}(gM_T).$$

It is an affine cocycle, $c(gh) = gc(h) + c(g)$, and its value span is G-stable because

$$b\,c(g) = c(bg) - c(b). \tag{3.4b}$$

Distinct matchings have distinct products: unique factorization recovers the normalized secant factors. Hence every nonbase orbit point gives a nonzero cocycle value. In type A_3, irreducibility of M_2 now forces the value span to be all of R_1.

In type B_3, project the cocycle to $M_0 = \chi$. Its restriction to G^+ is a homomorphism from the simple group $\mathrm{PSL}_2(7)$ to the additive group of $\mathbb{F}_7$, and is therefore zero. A nonbase point on the G^+-sheet consequently gives a nonzero vector in M_4; irreducibility and (3.4b) force the whole five-space $M_4 = \mathcal{H}_2$.

Lemma 3.5 supplies an outer-sheet value with nonzero radial projection. Since Δ_Q kills $M_4 = \mathcal{H}_2$, its value has nonzero $M_0 = \mathbb{F}_7 Q$-projection. The span already contains M_4, so it contains M_0 as well. Hence $W_{B_3} = R_2$, proving the first two rows of (3.4).

We prove the H_3 row equivariantly. Put $G = \mathrm{PGL}_2(11)$, $G^+ = \mathrm{PSL}_2(11)$, and let $\chi(g) = \det(g)^5$ be the square-class character. On the conic, the common restriction of the matching products is the Dickson form $s^{11}t - st^{11}$, of contragredient weight $\det^{-1}$. Let $V = \mathbb{F}_{11}^2$, with binary forms carrying the contragredient $\mathrm{GL}_2(11)$-action. The equivariant Fischer decomposition obtained from the Clebsch identification $\mathrm{Sym}^2 V \simeq \mathbb{F}_{11}^3$, together with the Dickson weight in the quotient transformation (2.2), gives the linearized module

$$\widetilde{R}_4 = R_4 \otimes \overset{-1}{\det} = M_8 \oplus M_4 \oplus M_0, \tag{3.5}$$

Explicitly, $g \in \mathrm{GL}_2(11)$ acts on the first expression by

$$g \cdot f = \det(g)^{-1}(f \circ \rho(g)^{-1}).$$

The three summands are

$$\begin{aligned} M_8 &= \mathcal{H}_4 \otimes \overset{-1}{\det} \simeq \mathrm{Sym}^8(V^\vee) \otimes \overset{-1}{\det}, \\ M_4 &= Q\mathcal{H}_2 \otimes \overset{-1}{\det} \simeq \mathrm{Sym}^4(V^\vee) \otimes \overset{-3}{\det}, \\ M_0 &= \mathbb{F}_{11} Q^2 \otimes \overset{-1}{\det} \simeq \overset{-5}{\det} = \chi. \end{aligned} \tag{3.6}$$

Equivalently, all three summands have the uniform form

$$M_r = \mathrm{Sym}^r(V^\vee) \otimes \overset{r/2}{\det} \otimes \chi, \qquad r = 8, 4, 0.$$

A scalar λI acts here as $\lambda^{-r}\lambda^r(\lambda^2)^5 = 1$, so these modules and their direct sum descend from $\mathrm{GL}_2(11)$ to G. This is the G-action used below. The determinant twist changes only the linearization, not the underlying Fischer subspaces $\mathcal{H}_4, Q\mathcal{H}_2, \mathbb{F}_{11}Q^2$ of R_4.

We need two elementary cohomology facts. Let $U = \langle u \rangle$ be the upper-unipotent Sylow 11-subgroup and let a generate the diagonal torus of order 10. Since $r < 10$, the cyclic norm $1 + u + \cdots + u^{10} = (u-1)^{10}$ vanishes on M_r, while $(u-1)M_r$ has codimension one. Thus $H^1(U, M_r)$ is represented by the class of x^r. Diagonal conjugation acts on that class by

$$a^{4-r/2}. \tag{3.7}$$

Restriction from G to U is injective because $[G:U]=120$ is invertible in $\mathbb{F}_{11}$, and its image is fixed by the Sylow normalizer. Equation (3.7) therefore gives

$$H^1(G,M_4)=H^1(G,M_0)=0, \qquad \dim H^1(G,M_8)\le 1. \tag{3.8}$$

The stabilizer $H=A_5$ of M_{H_3} lies in G^+. The restrictions of M_8, M_4, M_0 to H are respectively $4\oplus 5, 5, 1$, so

$$M_8^H=M_4^H=0, \qquad \dim M_0^H=1. \tag{3.9}$$

Equivalently, average the characters $(9,1,0,-1,-1)$, $(5,1,-1,0,0)$, and $(1,1,1,1,1)$ over the A_5 class sizes $(1,15,20,12,12)$.

Let $c(g)=\Phi_{M_{H_3}}(gM_{H_3})$. The common-restriction identity gives

$$c(gh)=gc(h)+c(g),$$

and c vanishes on H. Project c through (3.5). By (3.8), its M_4-component is a coboundary; its translating vector lies in $M_4^H=0$, so that component vanishes. This proves, without quotient coordinates,

$$\Delta_Q\Phi_{M_{H_3}}(M)\in\mathbb{F}_{11}Q \quad \text{for every } M\in\Omega_{H_3}. \tag{3.10}$$

Likewise the M_0-component is a coboundary and hence vanishes on G^+. Choose any other matching in the eleven-element G^+-orbit of M_{H_3}. Its quotient is nonzero: equality of the two secant products would identify their irreducible line factors and hence the matchings. Its M_4- and M_0-components vanish, so it gives a nonzero vector in M_8. The degree-eight restricted $\mathrm{SL}_2(\mathbb{F}_{11})$-module is irreducible [10, Fact 3.1]. Moreover the span of the cocycle values is G-stable, since

$$b\,c(g)=c(bg)-c(b) \qquad (b,g\in G).$$

Therefore the quotient span contains all of M_8.

Lemma 3.5 supplies an outer-sheet value with nonzero deepest radial projection. Since Δ_Q^2 kills $M_8\oplus M_4$, this supplies the M_0-projection. The span already contains M_8, so it also contains M_0. Hence $W_{H_3}=\mathcal{H}_4\oplus\mathbb{F}_{11}Q^2$, of dimension 10. Since $\dim R_4=15$, this equality is proper. $\square$

Remark 3.8 (Historical radial coordinates). Direct coordinates give $\Delta_Q\Phi=4$ for the displayed B_3 pair and $\Delta_Q^2\Phi=10$ for the displayed H_3 pair. These values independently check the two specializations of Lemma 3.5; they are not premises of its Dickson-recurrence proof.

Corollary 3.9 (The base-choice class and its homogeneous extension)**.** *Let $c_8: G\to M_8$ be the top-harmonic projection of the base-choice cocycle c. Then*

$$H^1(G,M_8)=\mathbb{F}_{11}[c_8]. \tag{3.11a}$$

In particular, the affine H_3 quotient configuration has no G-equivariant origin: no translation makes its affine action linear. Equivalently, the homogeneous module

$$E_{c_8}=M_8\oplus\mathbb{F}_{11}, \qquad g\cdot(v,t)=(gv+t\,c_8(g),t), \tag{3.11b}$$

is nonsplit. Up to equivalence and nonzero rescaling of the quotient coordinate, it is the unique nonsplit extension of the trivial G-module by M_8.

Proof. Equation (3.8) gives $\dim H^1(G, M_8) \leq 1$. On the base sheet the M_0-component vanishes, and the M_4-component vanishes by (3.9); the nonzero same-sheet quotient used in the proof above is therefore a value of c_8. If $c_8(g) = gv - v$, its vanishing on $H = \mathrm{Stab}_G(M_{H_3})$ would put v in $M_8^H = 0$, forcing $c_8 = 0$, a contradiction. Thus $[c_8] \neq 0$, proving (3.11a).

Changing the reference quotient translates the affine configuration and changes c_8 by a coboundary. Hence a G-equivariant origin would kill $[c_8]$, which is impossible. Finally the standard identification $\mathrm{Ext}^1_{\mathbb{F}_{11}G}(\mathbb{F}_{11}, M_8) \simeq H^1(G, M_8)$ identifies (3.11b) with $[c_8]$; the one-dimensionality just proved gives the last assertion. ☐

Corollary 3.10 (The exact H_3 loss)**.** *Over $\mathbb{F}_{11}$, the quartic form space has the Fischer decomposition*

$$R_4 = \mathcal{H}_4 \oplus Q\mathcal{H}_2 \oplus \mathbb{F}_{11}Q^2. \tag{3.12}$$

Consequently,

$$R_4 = W_{H_3} \oplus Q\mathcal{H}_2, \qquad R_4/W_{H_3} \simeq Q\mathcal{H}_2. \tag{3.13}$$

Equivalently,

$$W_{H_3} = \{f \in R_4 : \Delta_Q f \in \mathbb{F}_{11}Q\}. \tag{3.14}$$

Thus the H_3 quotient configuration retains the top harmonic nine-space and the deepest radial line, and omits precisely the five-dimensional middle radial layer.

Proof. For a homogeneous form f of degree r, direct differentiation gives

$$\Delta_Q(Qf) = (4r + 6)f + Q\Delta_Q f. \tag{3.15}$$

Hence $\Delta_Q(Qh) = 3h$ for $h \in \mathcal{H}_2$, while $\Delta_Q(Q^2) = 9Q$ over $\mathbb{F}_{11}$. Applying Δ_Q^2 and then Δ_Q shows that the three summands in (3.12) meet trivially. Their dimensions are $9, 5, 1$, so they exhaust the fifteen-dimensional space R_4. Theorem 3.7 then gives (3.13). If $f = h_4 + Qh_2 + cQ^2$ is its decomposition, then

$$\Delta_Q f = 3h_2 + 9cQ.$$

The quadratic decomposition $R_2 = \mathcal{H}_2 \oplus \mathbb{F}_{11}Q$ therefore gives (3.14). ☐

The rank proof has two distinct boundaries. The common restriction and quotient are general consequences of the conic ideal, and (3.3) describes the intrinsic target subspaces. For A_3, irreducibility alone forces the image. For B_3, irreducibility forces the top summand. For H_3, (3.7)–(3.10) prove the containment and top-summand nonvanishing equivariantly. Lemma 3.5 proves radial nonvanishing in both types from one recurrence. The primary certificate still enumerates every orbit and recovers the ranks, while an independent replay reconstructs that calculation and the common cycle mechanism.

This proof is uniform at the level of its mechanism—an affine connecting cocycle, its Sylow-normalizer weights, stabilizer fixed spaces, and irreducible summands—but it still uses the H_3-specific $A_5 < \mathrm{PGL}_2(11)$ module data. A theorem over all good reductions would additionally require integral models and an analysis of the characteristics in which the harmonic decomposition, orbit structure, or marker algebra degenerates.

Corollary 3.10 identifies the linear information lost by the H_3 orbit, but linear span does not recover the two factorization sheets inside the finite configuration. Their recovery begins one categorical level higher, with products of affine evaluation functions.

4 Balanced sheets and cubic orientation

For $T = B_3, H_3$, put $q = 7, 11$, respectively. The subgroup

$$G^+ = \mathrm{PSL}_2(q) \subset G = \mathrm{PGL}_2(q)$$

splits Ω_T into two orbits $\Omega_+ \sqcup \Omega_-$, each of size q; an element of the outer coset exchanges them. These are the two one-factorization sheets. The A_3 orbit does not split: its S_4 stabilizer cannot lie in $\mathrm{PSL}_2(5) \simeq A_5$, already by order.

Write $x_M = \Phi_{M_T}(M) \in W_T$. Let $L \subseteq \mathbb{F}_q^{\Omega_T}$ be the evaluation space of affine-linear functions on the finite configuration $\{x_M\}$. Since the points affinely span W_T,

$$\dim L = 1 + \dim W_T = q.$$

For subspaces of functions, write

$$L^{\circ 2} = \mathrm{span}\{fg : f, g \in L\},$$

where multiplication is pointwise. The common second-moment form in the proposition below is

$$B(\lambda, \mu) = \sum_{M \in \Omega_+} \lambda(x_M)\mu(x_M) = \sum_{M \in \Omega_-} \lambda(x_M)\mu(x_M)$$

on $W_T^\vee$. A *field-valued strength-two trade* is a weight $\tau \in k^\Omega$ satisfying, for all $f, g \in L$,

$$\sum_{M \in \Omega} \tau(M) f(M) g(M) = 0.$$

The B_3 model. Here the quotient configuration consists of fourteen points spanning the six-space W_{B_3}, split into two seven-point sheets. Its affine evaluation space L has dimension seven. Restriction to either sheet has rank six, while the common second-moment form has rank five and a radical functional that takes distinct constant values on the two sheets. It follows that $L^{\circ 2}$ is the thirteen-dimensional hyperplane of functions with equal sheet sums; its one-dimensional annihilator is the sheet sign. Thus the quadratic algebra recovers the 7+7 partition without a search through partitions. The first nonzero signed moment is cubic, so $L^{\circ 3} = \mathbb{F}_7^{14}$, and the homogenized points have h-vector $(1, 6, 6, 1)$. The next proposition isolates the quadratic step in a form that applies to both balanced configurations.

Proposition 4.1 (Radical–Hadamard recovery)**.** *Let k be a field and $\Omega = \Omega_+ \sqcup \Omega_-$, with both sheets of size $q = 0$ in k. Suppose an affine evaluation space $L \subseteq k^\Omega$ has the following properties:*

(i) *each sheet has zero first moment, and the two sheets have equal second moments;*

(ii) *$\dim L = q$, and restriction from L onto the zero-sum hyperplane of either sheet has rank $q - 1$;*

(iii) *the second-moment form on linear functionals has rank $q - 2$ and a one-dimensional radical;*

(iv) *a nonzero radical functional is constant on each sheet, with different constants on the two sheets.*

Then

$$L^{\circ 2} = \left\{ (u_+, u_-) : \sum_{\Omega_+} u_+ = \sum_{\Omega_-} u_- \right\}, \tag{4.1}$$

and

$$(L^{\circ 2})^{\perp} = k(\mathbf{1}_{\Omega_+} - \mathbf{1}_{\Omega_-}). \tag{4.2}$$

Consequently every field-valued strength-two trade is a scalar multiple of the sheet sign.

Proof. Let e_+ and e_- be the two sheet indicators. The radical functional evaluates as $a_+e_+ + a_-e_-$ with $a_+ \neq a_-$. Together with $1 = e_+ + e_- \in L$, it therefore yields $e_+, e_- \in L$.

Let $H_\pm$ be the zero-sum hyperplane on $\Omega_\pm$. The first-moment hypothesis and $q = 0$ put each restriction of L inside $H_\pm$; the rank hypothesis makes both restrictions surjective. Multiplication by e_+ and e_- now gives

$$(H_+ \oplus 0) + (0 \oplus H_-) \subseteq L^{\circ 2}. \tag{4.3}$$

This subspace has dimension $2q - 2$.

For $f, g \in L$, the difference between the two sheet sums of fg expands into constant, first-moment, and second-moment terms. The hypotheses make all three differences zero, so $L^{\circ 2}$ lies in the hyperplane on the right of (4.1). Finally choose restrictions $a, b \in H_+$ with nonzero standard pairing and lift them to $f, g \in L$. Equality of the second moments gives the same nonzero pairing on Ω_-. Thus fg has equal nonzero sheet sums and does not lie in (4.3). We have found $2q - 1$ independent directions in the $2q - 1$-dimensional hyperplane, proving (4.1). Its annihilator under the standard coordinate pairing is the line in (4.2). ☐

Proposition 4.2 (The modular-permutation mechanism)**.** *For $T = B_3, H_3$, the affine quotient configuration satisfies all four hypotheses of Proposition 4.1. More precisely, write*

$$W_T = U_T \oplus \mathbb{F}_q r, \qquad U_{B_3} = \mathcal{H}_2, \quad U_{H_3} = \mathcal{H}_4.$$

On either q-point sheet, affine evaluation has image the zero-sum hyperplane and rank $q - 1$. The common second-moment form has rank $q - 2$, with radical the radial functional $r^\vee$; that functional is constant on each sheet and its two constants are distinct.

Proof. As a G^+-module, U_T is the irreducible module M_{q-3} of dimension $q - 2$. Project the affine quotient points to U_T, writing the resulting sheet as $\{y_\omega\}$. The cocycle argument in Section 3 says that the differences $y_\omega - y_{\omega'}$ span U_T. Moreover $\sum_\omega y_\omega$ is G^+-fixed: translating the sheet adds q times the cocycle value, which vanishes in characteristic q. Since $U_T^{G^+} = 0$,

$$\sum_\omega y_\omega = 0.$$

Let $P = \mathbb{F}_q^{\Omega_+}$, and put

$$P_0 = \left\{ (a_\omega) : \sum_\omega a_\omega = 0 \right\}, \qquad S = \mathbb{F}_q \mathbf{1}.$$

Here $S \subset P_0$, the defining-characteristic feature that drives the argument. The map

$$P_0 \longrightarrow U_T, \qquad (a_\omega) \longmapsto \sum_\omega a_\omega y_\omega$$

is onto because the differences span, and the preceding moment identity kills S. Both P_0/S and U_T have dimension $q-2$, so it induces an isomorphism

$$P_0/S \simeq U_T.$$

The standard dot product on P_0 has radical exactly S. Dualizing therefore shows that the evaluations of $U_T^\vee$ form a nondegenerate $(q-2)$-space complementary to S in P_0. Adding constants gives all of P_0, proving the restriction rank and the top second-moment rank. The same argument applies to the other sheet.

The two top second moments are not merely proportional. On

$$M_{q-3} = \mathrm{Sym}^{q-3}(V^\vee) \otimes \overset{(q-3)/2}{\det} \otimes \chi$$

the standard apolar pairing is G-invariant: its determinant weight $\det^{-(q-3)}$ is cancelled by the square of the displayed twist, and $\chi^2 = 1$. Each sheet moment is a nonzero G^+-invariant form, hence a scalar multiple of this pairing, while an outer element transports one sheet to the other by an apolar isometry. Their scalars, and thus the two forms, are equal.

Finally, projection of the affine cocycle to the radial character χ restricts on the simple group G^+ to an additive homomorphism, hence vanishes. The radial coordinate is consequently zero on the base sheet and constant on the other. The secant-norm calculation in Lemma 3.5 shows that the outer constant is nonzero in both types. Because $q = 0$, radial–radial and radial–top second moments vanish. Thus $r^\vee$ is exactly the one-dimensional radical, and its sheet constants are distinct. □

For reference, exact reconstruction of the two configurations gives

T	q	sheet restriction ranks	second-moment rank	$\dim L^{\circ 2}$
B_3	7	$6, 6$	5	13,
H_3	11	$10, 10$	9	21.

(4.4)

In each row the radical is one-dimensional and takes two distinct sheet values. Proposition 4.2, rather than this table, proves these facts. The exact certificate and its independent reconstruction are cross-checks only; no enumeration of equal-size subsets or row reduction is used in the proof.

Lemma 4.3 (A full-support quadratic relation forces the cube)**.** *Let Ω be finite and let $L \subseteq k^\Omega$ be a unital linear space with* $\dim L > 1$. *Suppose*

$$L^{\circ 2} = \left\{ u : \sum_{\omega \in \Omega} \epsilon_\omega u_\omega = 0 \right\}$$

for some $\epsilon \in (k^\times)^\Omega$. Then

$$L^{\circ 3} = k^\Omega.$$

Proof. If a nonzero vector η annihilated $L^{\circ 3}$, then $L^{\circ 2} \subseteq L^{\circ 3}$ would put η in $k\epsilon$. For every $f \in L$, the vector $\eta \circ f$ also annihilates $L^{\circ 2}$, because

$$\langle \eta \circ f, h \rangle = \langle \eta, fh \rangle = 0 \qquad (h \in L^{\circ 2}).$$

Thus $\eta \circ f \in k\epsilon = k\eta$. Since η has full support, every $f \in L$ is constant, contrary to $\dim L > 1$. The annihilator of $L^{\circ 3}$ is therefore zero. □

Theorem 4.4 (Balanced sheets and cubic-first orientation)**.** *For $T = B_3, H_3$, the unordered pair $\{\Omega_+, \Omega_-\}$ is intrinsic to the affine quotient configuration:*

(i) *up to interchange, Ω_+ and Ω_- are the only complementary halves with equal first and second tensor moments;*

(ii) *if ϵ is $+1$ on Ω_+ and -1 on Ω_-, then*

$$\mu_j = \sum_{M\in\Omega_T} \epsilon(M) x_M^{\odot j} \in \mathrm{Sym}^j(W_T)$$

satisfies

$$\mu_1 = 0, \qquad \mu_2 = 0, \qquad \mu_3 \neq 0; \tag{4.5}$$

(iii) *μ_3 is independent of the reference matching, G^+ fixes it, and the outer coset negates it. Within G,*

$$\mathrm{Stab}_G(\mu_3) = G^+, \qquad \mathrm{Stab}_G(\mathbb{F}_q\mu_3) = G. \tag{4.6}$$

Thus degree three is the first signed tensor moment, or linear functional of such a moment, that orients the recovered sheet pair. This minimality statement concerns signed tensor moments, not every possible nonlinear statistic of the configuration.

Proof mode. Proposition 4.1 is a symbolic proof. Proposition 4.2 proves its hypotheses from the defining-characteristic permutation module and the shared alternating-cycle witness. Lemma 4.3 forces the nonzero cubic without a tensor calculation or Gorenstein premise. Signed Gale duality records self-association, while maximal-isotropic quotient duality gives the Gorenstein pairing. Exact finite arithmetic independently cross-checks the displayed ranks.

Proof. Equality of first and second tensor moments for a complementary pair is equivalent, by polarization, to its sign weight annihilating $L^{\circ 2}$. Since the characteristics are 7 and 11, polarization through degree two is valid. Proposition 4.1 says that the annihilator is exactly the sheet-sign line. A sign weight taking only the values $\{\pm 1\}$ is therefore ϵ or $-\epsilon$, proving (i).

The same moment equalities give $\mu_1 = \mu_2 = 0$. Proposition 4.1 identifies $L^{\circ 2} = \ker\langle \epsilon, -\rangle$, and ϵ has full support. Lemma 4.3 gives $L^{\circ 3} = \mathbb{F}_q^{\Omega_T}$. Hence ϵ cannot annihilate all cubic products. Since the lower signed moments vanish and 6 is invertible in $\mathbb{F}_7, \mathbb{F}_{11}$, polarization gives $\mu_3 \neq 0$.

For the geometric consequence, let A be the q-by-$2q$ matrix whose M-th column is $\widehat{x}_M = (1, x_M)$, and put $D = \mathrm{diag}(\epsilon(M))$. Equal sheet sizes and the vanishing signed moments through degree two give

$$ADA^{\mathsf{T}} = 0.$$

The affine spanning statement gives $\operatorname{rank} A = q$, so the rows of AD form the kernel of A. Thus AD is a Gale matrix, and its columns differ from those of A only by the nonzero scalars $\epsilon(M)$. The homogenized configuration is self-associated.

We now prove the Gorenstein statement directly. On

$$V = \mathbb{F}_q^{\Omega_+} \oplus \mathbb{F}_q^{\Omega_-}$$

use the nondegenerate signed coordinate form

$$B(u, v) = \sum_{\Omega_+} u_M v_M - \sum_{\Omega_-} u_M v_M. \tag{4.6a}$$

Proposition 4.1 identifies

$$L^{\circ 2} = \langle \mathbf{1} \rangle^{\perp_B}. \tag{4.6b}$$

For $u, v \in L$, the product $u \circ v$ lies in the right side of (4.6b), so $B(u, v) = 0$. Thus L is isotropic. Since $\dim L = q = \frac{1}{2} \dim V$, it is maximal isotropic:

$$L = L^{\perp_B}. \tag{4.6c}$$

Reduce the homogeneous coordinate ring by the homogenizing coordinate. Its positive graded pieces are

$$\mathcal{A}_1 = L/\langle \mathbf{1} \rangle, \qquad \mathcal{A}_2 = L^{\circ 2}/L, \qquad \mathcal{A}_3 = V/L^{\circ 2}.$$

The form B descends to

$$\mathcal{A}_1 \times \mathcal{A}_2 \longrightarrow \mathbb{F}_q. \tag{4.6d}$$

Its left radical is $(L^{\circ 2})^{\perp}_B/\langle \mathbf{1} \rangle = 0$, and its right radical is $L^{\perp}_B/L = 0$. Hence (4.6d) is perfect. Lemma 4.3 gives $L^{\circ 3} = V$, so $\mathcal{A}_3$ is one-dimensional and multiplication realizes (4.6d) as the degree-1-by-degree-2 socle pairing. Therefore $\mathcal{A}$ is Artinian Gorenstein with Hilbert function

$$(1, q-1, q-1, 1).$$

The homogenizing coordinate is regular because it is nonzero at every point, so the projective coordinate ring is arithmetically Gorenstein. The unique quadratic dependence is ϵ, which has full support; hence every one-point deletion imposes independent conditions on quadrics.

If the reference matching changes, every point translates by one vector c. Expanding the signed third moment after $x_M \mapsto x_M + c$ leaves μ_3 unchanged: the other terms contain μ_2, μ_1, or $\mu_0 = \sum_M \epsilon(M) = 0$.

Every element of G^+ preserves both sheets and hence fixes the signed sum. Every element of the outer coset exchanges the sheets and negates it. Since $\mu_3 \neq 0$, these two transformation laws give (4.6). Finally (4.5) proves the stated minimality inside the signed tensor-moment family. $\square$

Corollary 4.5 (Graded evaluation algebra)**.** *Put $L^{\circ 0} = \langle 1 \rangle$, and let $L^{\circ d}$ be the span of the pointwise products of d elements of L. For $T = B_3, H_3$, the pointwise powers of the affine evaluation space satisfy*

$$\dim L^{\circ d} = \begin{cases} 1, & d = 0, \\ q, & d = 1, \\ 2q-1, & d = 2, \\ 2q, & d \geq 3. \end{cases}$$

In particular, $L^{\circ 3} = \mathbb{F}_q^{\Omega_T}$. The homogenized configuration

$$\widehat{X}_T = \{[1 : x_M] : M \in \Omega_T\} \subseteq \mathbb{P}^{q-1}$$

therefore has Hilbert function

$$1,\ q,\ 2q-1,\ 2q,\ 2q, \ldots$$

and h-vector $(1, q-1, q-1, 1)$.

Proof. Pair functions on Ω_T with the sheet sign by $\langle \epsilon, f\rangle = \sum_M \epsilon(M) f(M)$. Proposition 4.1 gives $L^{\circ 2} = \ker\langle \epsilon, -\rangle$, with ϵ of full support. Lemma 4.3 gives $L^{\circ 3} = \mathbb{F}_q^{\Omega_T}$. Multiplication by 1 gives the same equality in every higher degree. Homogeneous degree-d forms on $\widehat{X}_T$ restrict to $L^{\circ d}$, which gives the Hilbert function; its first difference is the stated h-vector. Changing the reference matching translates every x_M. Such a translation preserves L and all its pointwise powers and acts projectively on $\widehat{X}_T$, so the graded algebra and Hilbert function are independent of the reference. $\square$

Corollary 4.6 (Self-association and cubic duality)**.** *For $T = B_3, H_3$, the configuration $\widehat{X}_T \subseteq \mathbb{P}^{q-1}$ is a reduced self-associated arithmetically Gorenstein set of $2q$ points. Every one-point deletion imposes independent conditions on quadrics, so $\widehat{X}_T$ has the Cayley–Bacharach property in degree two. Its dualizing residue vector is the sheet sign ϵ. The Artinian reduction by the homogenizing coordinate has Hilbert function $(1, q-1, q-1, 1)$, and its Macaulay inverse system is the cubic line $\mathbb{F}_q\mu_3$.*

Proof. The proof of Theorem 4.4 already establishes signed Gale self-duality, the Cayley–Bacharach property in degree two, the Gorenstein coordinate ring, and the displayed h-vector. These arguments commute with base change to an algebraic closure, so the Gorenstein conclusion descends to $\mathbb{F}_q$.

Let z be the homogenizing coordinate. It is nonzero at every point and is therefore regular on the coordinate ring. The degree d part of the reduction modulo z is $L^{\circ d}/L^{\circ(d-1)}$, so its Hilbert function is the h-vector from Corollary 4.5. The functional

$$f \longmapsto \sum_M \epsilon(M) f(x_M)$$

annihilates degrees below three and is nonzero in degree three. Thus ϵ is the socle residue vector. Degree-three polarization identifies its Macaulay dual generator with μ_3. $\square$

The implication above is the zero-defect case of the modern self-dual-code criterion: the Gorenstein defect is measured by the defect of the Schur square, or equivalently by decomposability [8, Theorem 3.11 and Corollaries 3.12–3.13]. Here $\dim L^{\circ 2} = 2q-1$ already places both configurations in the zero-defect case. The ideal, deletion, socle, inverse-system, and minimal-resolution calculations are also checked directly in the verification bundle. Self-association does not select an affine origin. Translating all x_M acts projectively on the columns of A and preserves the signed Gale identity, in agreement with the nontrivial base-choice obstruction behind the quotient construction.

The cubic remembers an orientation that the linear quotient does not. Indeed, the signed linear relation

$$\sum_{M\in\Omega_+} x_M = \sum_{M\in\Omega_-} x_M$$

already shows that the sheet sign is killed by the linear quotient map. Quadratic products recover the unordered partition through (4.2), and the cubic supplies the sign exchanged when the two recovered levels are swapped.

Corollary 4.7 (Secant-product tensor syzygies)**.** *For $T = B_3, H_3$, let $m_Q : R_{m-2} \to R_m$ denote multiplication by Q. Regarding the displayed powers as symmetric tensors, not* *polynomial powers, the canonically scaled secant products satisfy*

$$\begin{aligned}\sum_M \epsilon(M)P_M &= 0,\\ \sum_M \epsilon(M)P_M^{\odot 2} &= 0,\\ \sum_M \epsilon(M)P_M^{\odot 3} &= \mathrm{Sym}^3(m_Q)(\mu_3) \neq 0.\end{aligned} \tag{4.7}$$

Proof. Substitute $P_M = P_{M_T} + Qx_M$ and expand. Each sheet has $q = 0$ elements in $\mathbb{F}_q$, and the signed quotient moments vanish through degree two. All terms below cubic degree cancel, leaving $\mathrm{Sym}^3(m_Q)(\mu_3)$ in degree three. Since $\mathbb{F}_q[X, Y, Z]$ is a domain, multiplication by Q is injective; over a field, the symmetric powers of an injective linear map are injective. □

Conclusion

Restriction to a conic erases the pairing of marked points, but division by its equation retains a structured record of the factorization. The four-endpoint switch produces an affine quotient configuration, and the three rank-three systems span spaces of dimensions $3, 6, 10$. In type H_3, the quotient retains the top harmonic part and deepest radial line while losing exactly $Q\mathcal{H}_2$; its base-choice cocycle also obstructs an equivariant affine origin.

In the balanced B_3 and H_3 cases, quadratic products recover the unordered factorization sheets, and their first signed tensor moment is cubic. The same cubic is the Macaulay inverse system of the self-associated arithmetically Gorenstein homogenization. Thus the information forgotten by conic restriction returns in two stages: quadratic data recover the unoriented factorization, and cubic data recover its orientation.

The affine-origin obstruction is not the end of the cubic. Once the icosahedral marking is fixed, its orientation line survives in a cross-characteristic lift.

The completeness theorem shows that, among full projective matching orbits, a one-dimensional strength-two trade space with a two-valued generator occurs only for $B_3/\mathbb{F}_7$ and $H_3/\mathbb{F}_{11}$. The proof derives the one-factorization sheets rather than assuming them. Its extension-field input is one Frobenius-parity bit and one local first-wall spill, and the two radial projections are specializations of one edge-selected alternating-cycle recurrence.

The appendices record further consequences and refinements of the H_3 configuration that are not needed for the main theorem. They require a chosen A_4-structure and do not canonically identify the depth plane with the relative-cubic plane. Appendices A–C treat decorated incidence profiles, their modular depth, and arithmetic marker splitting; Appendices D and E isolate the resulting cubic structures. The final appendix gives the verification architecture for all statements.

A Six profiles and matching-row reconstruction

We now specialize to H_3. Put

$$G = \mathrm{PGL}_2(11), \qquad G^+ = \mathrm{PSL}_2(11), \qquad X = \Omega_{H_3} \simeq G/H,$$

where $H \simeq A_5$ is the stabilizer of the base matching in (3.1). Thus $X = X_+ \sqcup X_-$, with $|X_+| = |X_-| = 11$. The construction has four stages:

$$(A_+, A_-) \longmapsto K \longmapsto \text{four oriented cell pairs} \longmapsto D(M) \longmapsto K\backslash G/H.$$

The ordered parent pair supplies the A_4-subgroup K; its cell partition supplies four signed incidence counts; those counts recover the double-coset label. Only the first two stages require coordinates.

A.1 The decorated parent correspondence

On $V = \mathbb{F}_{11}^3$, set

$$q_0(x, y, z) = x^2 + y^2 + z^2, \qquad \mathcal{C}_0 = \mathbb{P}\{q_0 = 0\}.$$

The projectivity

$$\psi = \begin{pmatrix} 10 & 2 & 1 \\ 1 & 2 & 10 \\ 3 & 0 & 3 \end{pmatrix} \quad \text{satisfies} \quad q_0(\psi(X, Y, Z)^t) = 3(XZ - Y^2), \tag{A.1}$$

so $\psi \circ \nu$ identifies the endpoint conic of Section 2 with $\mathcal{C}_0$. In these coordinates take the two six-point sets

$$\begin{aligned} A_+ &= \{(0:1:4), (0:1:7), (1:0:3), (1:0:8), (1:4:0), (1:7:0)\}, \\ A_- &= \{(0:1:3), (0:1:8), (1:0:4), (1:0:7), (1:3:0), (1:8:0)\}. \end{aligned} \tag{A.2}$$

Let $\mathcal{P} = G \cdot A_+$, where G is viewed as the full projective stabilizer of $\mathcal{C}_0$. For $A \in \mathcal{P}$ and $a \in A$, let $C_{A,a}$ be the conic through $A \setminus \{a\}$. The six pairs

$$C_{A,a} \cap \mathcal{C}_0, \qquad a \in A, \tag{A.3}$$

form a perfect matching M_A of the twelve points of $\mathcal{C}_0$. On this 22-point orbit, $A \mapsto M_A$ is a G-equivariant bijection onto X. This is the *matching-decorated parent correspondence*; its bijectivity and the assertion that (A.3) consists of six disjoint pairs are exact finite inputs. It is the secant-factorization counterpart of the self-polar matching and Brianchon-support reconstruction obtained from the Clebsch decoding geometry [9].

Put $H_\pm = \mathrm{Stab}_G(A_\pm)$ and

$$K = H_+ \cap H_-, \qquad J = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}. \tag{A.4}$$

Then $K \simeq A_4$, $J(A_+) = A_-$, and J normalizes K and exchanges X_+ with X_-. Thus the *ordered golden pair* means precisely the ordered pair (A_+, A_-).

Choose linear representatives of K preserving q_0, and let $\widetilde{K} = \mathbb{F}_{11}^\times K$. For the following eight vectors, define $R_i = \mathbb{P}(\widetilde{K} r_i)$:

$$\begin{array}{cccc} i & r_i & i & r_i \\ \hline 1 & (0,1,4) & 10 & (0,1,3) \\ 3 & (0,1,2) & 13 & (0,1,5) \\ 6 & (1,3,2) & 14 & (1,1,5) \\ 9 & (1,4,2) & 11 & (1,2,4). \end{array} \tag{A.5}$$

These are cells in the scalar-K orbit partition of $\mathbb{P}(V)$. The involution J exchanges the four pairs

$$(R_1, R_{10}), \quad (R_3, R_{13}), \quad (R_6, R_{14}), \quad (R_9, R_{11}). \tag{A.6}$$

Only these four oriented pairs enter the argument. Reversing the golden pair reverses all four orientations.

This profile layer uses more decoration than the balanced-sheet theorem. The quotient configuration intrinsically recovers the unordered sheets, but it does not select an ordered golden pair or the resulting scalar-K refinement. The reconstruction below is relative to that selected refinement.

A.2 Incidence profiles

For $M \in X$, let S_M be the union of its six secants in $\mathbb{P}^2(\mathbb{F}_{11})$, and put

$$Z_M(R) = |S_M \cap R|.$$

Equivalently, $Z_M(R)$ counts the points of R at which P_M vanishes. Define the *depth profile*

$$\begin{aligned} D(M) = (&Z_M(R_1) - Z_M(R_{10}),\ Z_M(R_3) - Z_M(R_{13}), \\ &Z_M(R_6) - Z_M(R_{14}),\ Z_M(R_9) - Z_M(R_{11})) \in \mathbb{F}_{11}^4. \end{aligned} \tag{A.7}$$

The entries here and in the profile table below are first formed as integer differences of cardinalities, using the canonical cell orientation in (A.6), and only then reduced modulo 11. This definition uses only zero sets of secant products, so it is unchanged if their equations are rescaled.

Theorem A.1 (Six-profile reconstruction)**.** *The K-orbits on each sheet $X_\pm$ have sizes* $1, 4, 6$*. With the orientation in (A.6), the corresponding profiles and fibre sizes are*

sheet	$\lvert D^{-1}(v)\rvert$	v
$+$	1	$v_1 = (-6, 0, 12, -12),$
$+$	4	$v_2 = (-3, 3, 0, 3),$
$+$	6	$v_3 = (3, -2, -2, 0),$
$-$	1	$-v_1,$
$-$	4	$-v_2,$
$-$	6	$-v_3.$

(A.8)

The six vectors are pairwise distinct. Hence D recovers the double-coset row in

$$K\backslash G/H$$

of every matching. Its linear span nevertheless has dimension only two: over $\mathbb{F}_{11}$ it is the plane

$$2a + 2b + c = 0, \qquad 9a + 8b + d = 0. \tag{A.9}$$

The two singleton profiles recover their individual matchings. Under the matching-decorated parent correspondence, they therefore recover the corresponding individual H_3 parent. A profile in a fibre of size four or six recovers only that K-orbit, not an individual matching.

Proof mode. The subgroup-mark reduction and the implications from equivariance are conceptual. The six representative incidence rows in (A.10) below are certificate-checked exact counts and are reconstructed by an independent implementation. A companion certificate verifies that the matching decoration uniquely recovers its parent on the 22-point H_3 locus.

Proof. The permutation character of K on either eleven-point sheet has values

$$(11, 3, 2)$$

on elements of orders $1, 2, 3$. The marks of the transitive A_4-sets are

K-set	size	marks on orders $1, 2, 3$
K/K	1	$(1, 1, 1)$
K/V_4	3	$(3, 3, 0)$
K/C_3	4	$(4, 0, 1)$
K/C_2	6	$(6, 2, 0)$
$K/1$	12	$(12, 0, 0).$

Comparing marks gives the unique nonnegative decomposition

$$X_+|_K \simeq K/K \ \sqcup \ K/C_3 \ \sqcup \ K/C_2.$$

The orbit sizes are therefore $1, 4, 6$; conjugation by J gives the same decomposition on X_-.

For $k \in K$, the transformation k preserves every cell in (A.6) and carries S_M to S_{kM}. Thus $D(kM) = D(M)$. The involution J exchanges the two cells in every oriented pair and carries S_M to S_{JM}, so

$$D(JM) = -D(M).$$

It remains to count one representative on each positive-sheet orbit. The exact incidence rows are

orbit size	$\mathrm{Stab}_K(M)$	zero counts in the four pairs of (A.6)	$D(M)$
1	A_4	$(0,6;\ 0,0;\ 12,0;\ 0,12)$	v_1,
4	C_3	$(3,6;\ 3,0;\ 3,3;\ 6,3)$	v_2,
6	C_2	$(3,0;\ 1,3;\ 2,4;\ 6,6)$	v_3.

(A.10)

The J-images supply the three negative rows. These six vectors are pairwise distinct, so their fibres are precisely the six K-orbits.

Row reduction over $\mathbb{F}_{11}$ gives rank two and the equations (A.9). This rank drop does not merge orbit labels: linear rank and set-theoretic separation are different assertions. Finally, a singleton fibre contains its matching without residual choice. The stabilizer of the matching M_A from (A.3) is the same A_5 as the stabilizer of its parent A. The equivariant decorated transform between the two G/H-orbits is therefore a bijection, so the matching also determines its parent. The stated limitation for the other four rows follows from their fibre sizes. ☐

Corollary A.2 (Decorated sheet classifier). *Relative to the orientation in (A.6), the profile $D(M)$ determines the sheet containing M. The antipodal singleton pair $\{v_1, -v_1\}$ recovers the unordered golden matching pair and hence the unordered golden parent pair; choosing the orientation selects one member.*

Proof. The positive and negative profile sets in (A.8) are disjoint, and the only singleton fibres are v_1 and $-v_1$. Apply the matching–parent bijection from Theorem A.1. ☐

This pointwise classifier does not contradict the failure of a linear sheet sign in Section 4. The map D uses zero incidences with the selected scalar-K refinement; it is not a linear functional on the affine quotient configuration. Without that extra decoration, the quadratic and cubic moment statements remain the intrinsic recovery mechanism.

The profile compression also retains the cubic-first orientation from Section 4. The three positive vectors satisfy the integral weighted relation

$$v_1 + 4v_2 + 6v_3 = 0. \tag{A.11}$$

Corollary A.3 (Orbit weights from normalized profiles). *The three incidence-normalized vectors v_1, v_2, v_3 recover the orbit-size multiset $\{1, 4, 6\}$. They also recover the stabilizer orders $\{12, 3, 2\}$ inside $K \simeq A_4$. The underlying rational rays alone do not recover these weights.*

Proof. The coordinates of each v_i are fixed by the cardinality differences in (A.7), so independent rescaling is not allowed. These three integral vectors span a plane over $\mathbb{Q}$, and (A.11) is the primitive positive generator of their relation lattice. Its coefficients recover the three orbit sizes without labels. Orbit–stabilizer then gives $12/1, 12/4, 12/6$. If the vectors are replaced by primitive generators of their rays, the relation becomes $1 : 2 : 1$, which proves the final warning. ☐

B Modular depth and the projective-cover bridge

Because an H_3 sheet has eleven points in characteristic eleven, the sum of its basis vectors is simultaneously constant and augmentation-zero. Linearizing the three incidence rows therefore kills exactly this constant direction and leaves the two-dimensional profile plane. The following module statement identifies that loss.

Retain the notation of Section A, put

$$k = \mathbb{F}_{11}, \qquad L = G^+ = \mathrm{PSL}_2(11), \qquad V = k[X_+] \simeq \mathrm{Ind}_H^L \mathbf{1},$$

and regard V as the permutation module on one eleven-point sheet. Let e_1, e_4, e_6 be the indicator functions of the three K-orbits, in orbit-size order. Thus

$$V^K = ke_1 \oplus ke_4 \oplus ke_6, \qquad \mathbf{1} = e_1 + e_4 + e_6. \tag{B.1}$$

Linearizing the incidence construction in (A.7) gives

$$\mathcal{D} : V^K \longrightarrow k^4, \qquad e_i \longmapsto i\, u_i, \tag{B.2}$$

where $u_1 = v_1$, $u_4 = v_2$, and $u_6 = v_3$ are the profiles on the orbits of the indicated sizes. Equation (A.11) says exactly that $\mathcal{D}(\mathbf{1}) = 0$.

Proposition B.1 (Modular depth quotient)**.** *The L-module V is the projective cover $P(\mathbf{1})$ and has Loewy dimensions*

$$1 \mid 9 \mid 1.$$

The map $\mathcal{D}$ has kernel $k\mathbf{1}$ and induces an isomorphism

$$P(\mathbf{1})^K / \operatorname{soc} P(\mathbf{1}) \xrightarrow{\ \sim\ } \langle u_1, u_4, u_6 \rangle_k. \tag{B.3}$$

The sheet exchanged by J carries the negative copy of this quotient.

Proof mode. Projectivity, indecomposability, fixed-point exactness, and the socle quotient are conceptual. Absolute irreducibility of the nine-dimensional middle layer and the rank and kernel of $\mathcal{D}$ are exact generator-matrix calculations in the `relative-cubic-depth` bundle, reconstructed by its independent replay.

Proof. The order of $H \simeq A_5$ is prime to 11, so its trivial module is projective and induction makes V projective. The action of L on X_+ is doubly transitive. Hence $\mathrm{End}_L(V)$ is generated by the identity and the all-ones operator E. In characteristic eleven,

$$E^2 = 11E = 0,$$

so this endomorphism algebra is local. Thus V is indecomposable. Its trivial head occurs once, and consequently $V = P(\mathbf{1})$. The constant vector is its one-dimensional socle. The remaining nine-dimensional layer is absolutely irreducible; equivalently, the exact generator matrices on it span the full 9-by-9 matrix algebra. This gives the displayed Loewy dimensions.

Taking K-fixed points is exact because $11 \nmid |K| = 12$. The orbit sizes $1, 4, 6$ give (B.1). The three vectors in (B.2) span the plane (A.9), and their only relation is (A.11), so $\ker \mathcal{D} = k\mathbf{1}$. This proves (B.3). Finally $D(JM) = -D(M)$, which attaches the outer character on the opposite sheet. □

Corollary B.2 (The canonical nine-space bridge). *For the fixed marked conic and sheet X_+, there is a canonical G^+-module isomorphism*

$$\operatorname{rad} P(\mathbf{1})/\operatorname{soc} P(\mathbf{1}) \xrightarrow{\sim} M_8{\downarrow_{G^+}} . \tag{B.4}$$

It sends the augmentation class of $\sum_{M\in X_+} a_M[M]$ to

$$\sum_{M\in X_+} a_M\, \pi_8(x_M), \qquad \sum_M a_M = 0, \tag{B.5}$$

where π_8 is the top-harmonic projection. Formula (B.5) is independent of the reference matching.

Proof. The unique trivial head map in Proposition B.1 is the augmentation map, so

$$\operatorname{rad} P(\mathbf{1}) = \ker(V \xrightarrow{\sum} k).$$

Changing the reference matching translates every x_M by the same vector, so the augmentation condition makes (B.5) reference-independent. The affine cocycle term in x_{gM} cancels for the same reason, hence the map is G^+-equivariant. The same-sheet first-moment identity of Proposition 4.1 sends the constant vector $\mathbf{1}$, which lies in the augmentation module because $|X_+| = 11 = 0$ in k, to zero. Thus (B.5) descends to $\operatorname{rad} P(\mathbf{1})/\operatorname{soc} P(\mathbf{1})$.

Its image contains a nonzero same-sheet quotient and is G^+-stable. The irreducibility of $M_8{\downarrow_{G^+}}$ therefore makes it surjective. Both sides have dimension nine, so it is an isomorphism. □

Corollary B.3 (The homogeneous projective-cover bridge). *For the cocycle representative c_8 determined by the chosen reference matching, let E_{c_8} be the G-module defined in (3.11b). There is a G^+-module isomorphism*

$$P(\mathbf{1})/\operatorname{soc} P(\mathbf{1}) \xrightarrow{\sim} E_{c_8}{\downarrow_{G^+}}, \tag{B.6}$$

given by

$$\Theta\left(\sum_{M\in X_+} a_M[M]\right) = \left(\sum_{M\in X_+} a_M \pi_8(x_M),\ \sum_{M\in X_+} a_M\right). \tag{B.7}$$

Thus the extension class, unlike a cocycle representative or an affine origin, is canonical: the top-harmonic homogeneous extension restricts to the nonsplit exact sequence

$$0 \longrightarrow M_8{\downarrow_{G^+}} \longrightarrow P(\mathbf{1})/\operatorname{soc} P(\mathbf{1}) \longrightarrow \mathbf{1} \longrightarrow 0.$$

If $\widetilde{E}_c$ denotes the homogenization of the full affine quotient with translation space $M_8 \oplus M_0$, then

$$\widetilde{E}_c{\downarrow_{G^+}} \simeq P(\mathbf{1})/\operatorname{soc} P(\mathbf{1}) \oplus \mathbf{1}, \tag{B.8}$$

where the last summand is the radial module $M_0{\downarrow_{G^+}}$.

Proof. The affine transformation law $\pi_8(x_{gM}) = g\pi_8(x_M) + c_8(g)$ makes Θ G^+-equivariant. The constant vector maps to zero: its second coordinate is $|X_+| = 11 = 0$ in k, and its first coordinate vanishes by the same-sheet first-moment identity. Hence Θ descends through the socle.

On the kernel of augmentation, the descended map is the isomorphism of Corollary B.2; on the one-dimensional head it induces the identity through the second coordinate. It is therefore an isomorphism. The exact sequence cannot split: Proposition B.1 gives $P(\mathbf{1})/\operatorname{soc} P(\mathbf{1})$ the simple socle M_8, whereas a splitting would put a trivial summand in its socle. Finally, $M_0 = \chi$ is trivial on G^+, and its cocycle vanishes there, proving (B.8). □

Remark B.4 (Ambient and intrinsic sheet recovery). With the harmonic–radial decomposition retained, the M_0-projection already separates the two H_3 sheets: it is zero on X_+, constant on X_-, and nonzero by Lemma 3.5. The quadratic theorem of Section 4 is stronger in a different sense: it recovers the sheets intrinsically from the abstract affine point configuration, without retaining the ambient Fischer operator.

This quotient identifies the precise modular loss in the six-profile construction. Before linearization, the six profiles remain distinct and recover all six double-coset labels. After linearization, the constant orbit trade dies and only the two-dimensional depth plane remains. Conversely, the three rational rays in that plane remain distinct but do not retain their orbit weights. The weights $1, 4, 6$ are recovered only when the quotient is supplied with the incidence-normalized vectors of (A.8), as in Corollary A.3.

Remark B.5 (Reduction of the profile table). The four-coordinate profiles in (A.8) are integral before reduction. The greatest common divisor of their nonzero 2-by-2 minors is 3. Their span therefore drops below rank two only in characteristic 3. The six signed rows remain distinct away from characteristics 2 and 3: characteristic 2 identifies antipodes, while characteristic 3 sends the rows v_1 and v_2 to zero. This is a statement about the displayed integral table, not a construction of the marked conic geometry in those characteristics.

C Arithmetic splitting of the marker algebra

The arithmetic representatives of the three configurations come in conjugate pairs. Besides the base matchings in (3.1), we use

$$\begin{aligned} M_{B,-} &= \{\{0,\infty\},\{1,3\},\{2,6\},\{4,5\}\}, \\ M_{B,+} &= \{\{0,\infty\},\{1,5\},\{2,3\},\{4,6\}\}, \\ M_{H,-} &= \{\{0,10\},\{1,\infty\},\{2,7\},\{3,5\},\{4,8\},\{6,9\}\}. \end{aligned} \tag{C.1}$$

The pair for A_3 consists of two copies of M_{A_3}, the pair for B_3 is $(M_{B,-}, M_{B,+})$, and the pair for H_3 is $(M_{H_3}, M_{H,-})$. To make the arithmetic bridge part of the definition, let $\bar{k}$ be an algebraic closure of the working field and define the root-specialization maps

T	root	$\sigma_T(\text{root})$
A_3	$\sqrt{2},\ -\sqrt{2}$	$M_{A_3},\ M_{A_3}$
B_3	$3,\ 4$	$M_{B,-},\ M_{B,+}$
H_3	$4,\ 8$	$M_{H_3},\ M_{H,-}$.

(C.2)

Thus the *marker datum* is the quadratic algebra together with σ_T, not the polynomial alone. Exact projective transport places the displayed B_3 and H_3 values in Ω_{B_3} and Ω_{H_3}. The certificate also verifies that σ_T is the reduction of the corresponding integral Coxeter frames; the theorem below needs only the explicit map (C.2). We call the datum *fused* when its two geometric roots have one rational matching value, and *split* when they give two distinct rational matching sheets.

Lemma C.1 (Split–inert frame calculation)**.** *The three marker polynomials factor as follows:*

$$\begin{array}{lcc} A_3,\ q=5 & t^2-2 & \textit{irreducible}, \\ B_3,\ q=7 & t^2-2=(t-3)(t-4) & \textit{split}, \\ H_3,\ q=11 & t^2-t-1=(t-4)(t-8) & \textit{split}. \end{array} \tag{C.3}$$

In the first row the two geometric roots form one Frobenius orbit over $\mathbb{F}_{25}$. In each split row the nontrivial involution of the quadratic marker algebra exchanges the two rational roots.

Proof. The nonzero squares modulo 5 are 1 and 4, so 2 is not a square. If $u^2 = 2$ in $\mathbb{F}_{25}$, the nontrivial $\mathbb{F}_5$-automorphism sends u to $-u$, and hence exchanges the two roots. The remaining factorizations follow by direct substitution. $\square$

Theorem C.2 (Marker specialization and H_3 arithmetic gluing)**.** *For these arithmetic representatives of the rank-three matching configurations, the following hold.*

(a) *The A_3 marker datum is fused over $\mathbb{F}_5$. The B_3 and H_3 marker data split into two rational matching sheets over $\mathbb{F}_7$ and $\mathbb{F}_{11}$, respectively.*

(b) *In the H_3 case, the stabilizers H_+ and H_- of the singleton matching in each rational sheet are isomorphic to A_5, satisfy*

$$H_+ \cap H_- = K \simeq A_4, \qquad \langle H_+, H_- \rangle = \mathrm{PSL}_2(11), \tag{C.4}$$

and are exchanged by a projectivity J of nonsquare determinant. The normalizer $N_G(K)$ is a rational S_4 whose determinant-square kernel is K.

(c) *The determinant character that distinguishes the two rational sheets also negates the depth profiles. Thus choosing one root in the split marker algebra chooses one sheet, while its singleton profile chooses the corresponding decorated matching row. Forgetting that choice leaves the unordered pair.*

Proof mode. Lemma C.1 and the determinant-character deductions are conceptual. The agreement of the specialization map (C.2) with the integral Coxeter frames, the matching stabilizers and intersections, and the generated subgroup in (C.4) are exhaustive exact finite checks. The classical subgroup names use the standard subgroup structure of $\mathrm{PGL}_2(11)$ [1, 4].

Proof. Part (a) records the specialization data in (C.2). Its two A_3 roots have the same matching value. The two values in the B_3 row are exchanged by $x \mapsto -x$, and the two values in the H_3 row are exchanged by

$$x \longmapsto \frac{x-1}{x+1}.$$

The exact transport check verifies these identities on the displayed matchings. Together with Lemma C.1, they prove (a).

For H_3, exact projective closure gives

$$|H_+| = |H_-| = 60, \qquad |H_+ \cap H_-| = 12, \qquad |\langle H_+, H_- \rangle| = 660.$$

Both stabilizers lie in the determinant-square subgroup, while the displayed transporter has nonsquare determinant. The normalizer $N_G(K)$ has order 24, and its determinant-square kernel has order 12. The cited subgroup identifications give (C.4) and the S_4/A_4 hinge, proving (b).

The determinant-square subgroup has the two matching sheets as its orbits, and J exchanges them. Section A proves $D(JM) = -D(M)$ and identifies the unique singleton in each sheet. Theorem A.1 then returns the decorated matching row from that singleton. This proves (c). $\square$

Arithmetic splitting and modular depth retain different amounts of information. Splitting produces a two-point choice of sheet. The depth quotient is defined only after the A_4-refinement has been selected and remembers all three orbit rays modulo the constant socle. The first choice orients the second, but does not canonically identify it with any cubic quotient of the ten-dimensional factorization space. Appendix E makes that boundary precise.

D Cubic structure of the profile plane

Lemma D.1 (Three-ray cubic forcing)**.** *Let k have characteristic different from 2 and 3. Suppose u_1, u_2 form a basis of a two-dimensional k-space and $n_1, n_2, n_3 \in k^\times$ satisfy*

$$n_1 u_1 + n_2 u_2 + n_3 u_3 = 0.$$

For the signed antipodal measure

$$\xi = \sum_{i=1}^{3} n_i(\delta_{u_i} - \delta_{-u_i}),$$

the moments of degrees one and two vanish, while the cubic moment is nonzero.

Proof. The relation kills the first moment, and antipodality kills every even moment. Write

$$u_3 = -\frac{n_1}{n_3}u_1 - \frac{n_2}{n_3}u_2.$$

In $\sum_i n_i u_i^{\odot 3}$, the coefficient of $u_1^{\odot 2} \odot u_2$ is

$$-\frac{3n_1^2 n_2}{n_3^2},$$

which is nonzero. The signed cubic is twice this tensor and is therefore nonzero as well. □

Corollary D.2 (The mass-zero cubic flag)**.** *In Lemma D.1, suppose also that $n_1+n_2+n_3 = 0$. Then its half-cubic factors as*

$$\frac{1}{2}M_3(\xi) = \sum_{i=1}^{3} n_i u_i^{\odot 3} = \frac{n_1 n_2}{(n_1+n_2)^2}(u_1 - u_2)^{\odot 2} \odot ((2n_1+n_2)u_1 + (n_1+2n_2)u_2). \tag{D.1}$$

The two displayed lines are distinct. Hence the cubic intrinsically defines a doubled line and a residual simple line; its Hessian recovers the doubled line.

Proof. Since $n_3 = -(n_1+n_2)$, the weighted vector relation gives

$$u_3 = \frac{n_1 u_1 + n_2 u_2}{n_1 + n_2}.$$

Substitution and expansion give (D.1). The residual factor would equal the doubled factor only if $3(n_1+n_2) = 0$, which is impossible under the hypotheses. The Hessian of a binary cubic $L^2 R$, with L and R distinct, is a nonzero scalar multiple of L^2. □

For the incidence-normalized profiles of Section A, put

$$\eta = \delta_{v_1} + 4\delta_{v_2} + 6\delta_{v_3} - \delta_{-v_1} - 4\delta_{-v_2} - 6\delta_{-v_3}.$$

Then

$$M_j(\eta) = (1-(-1)^j)\sum_{i=1}^{3} n_i v_i^{\odot j}, \qquad (n_1, n_2, n_3) = (1, 4, 6). \tag{D.2}$$

Equation (A.11) kills the first moment, and antipodality kills every even moment. The first two profiles are independent over $\mathbb{F}_{11}$, since the determinant of their first two coordinates is $-18 \neq 0$. Lemma D.1 therefore forces the cubic to be nonzero. In the frozen normalization, its first coordinate is

$$2((-6)^3 + 4(-3)^3 + 6(3)^3) = 6 \neq 0 \quad \text{in } \mathbb{F}_{11}. \tag{D.3}$$

Thus the six-row quotient preserves the cubic-first signed moment even though its linear image is only a plane.

Remark D.3 (The H_3 cubic flag). Here $1 + 4 + 6 = 0$ in $\mathbb{F}_{11}$, so Corollary D.2 forces the double-line factorization. In the basis $e_1 = v_2, e_2 = v_3$ of the profile plane, the projective cubic is

$$f(x, y) = x^3 + 7x^2y + 5xy^2 + 9y^3 = (x - y)^2(x - 2y),$$

with Hessian $7x^2 + 8xy + 7y^2 = 7(x - y)^2$. Hence the cubic intrinsically distinguishes a doubled line and a residual simple line. This flag is internal to the decorated profile plane: it supplies no canonical map from the full relative-cubic space and no descended G^+-action.

E The relative-cubic Tate plane

Let $W = W_{H_3}$ be the ten-dimensional quotient space from Theorem 3.7, let χ be the determinant character of $G = \mathrm{PGL}_2(11)$, and put

$$M = \mathrm{Sym}^3(W) \otimes \chi, \qquad R = M^G.$$

The signed cubic of Section 4 lies in R, which has dimension three. There is a canonical two-dimensional quotient of this source space, but it is not canonically the depth plane of Proposition B.1.

We fix the monomial basis of $\mathrm{Sym}^3(W)$ induced by the harmonic–radial coordinates of Section 3. The *frozen relative basis* is the reduced-echelon basis of the simultaneous equations $gf = \chi(g)f$ for the standard generators defining the marked H_3 orbit, ordered by its pivots. Thus the coordinates below specify a reproducible basis, not an additional intrinsic structure. The *three labelled cubic orbit classes* are indexed, in order, by the three K-orbits of sizes $1, 4, 6$ on X_+. Finally, the *invariant symmetric-form pencil* is $\mathbb{P}((\mathrm{Sym}^2 W)^G)$. Its unique rank-one point has the form $[\ell^2]$; contraction by its covector ℓ is the induced map

$$\iota_\ell : R \longrightarrow (\mathrm{Sym}^2 W)^G.$$

Proposition E.1 (Canonical relative-cubic Tate plane)**.** *The invariant space R and the coinvariant space M_G both have dimension three. The canonical projection and group norm fit into*

$$R \xrightarrow{\pi} M_G \xrightarrow{N} R, \qquad \mathrm{rank}\, \pi = 2, \quad \mathrm{rank}\, N = 1, \tag{E.1}$$

with

$$\mathrm{im}\, \pi = \ker N, \qquad \mathrm{im}\, N = \ker \pi = \langle [1 : 3 : 9] \rangle. \tag{E.2}$$

Contraction by the covector whose square is the unique rank-one member of the invariant symmetric-form pencil gives a second construction of the same quotient. In the frozen relative basis its matrix is

$$\begin{pmatrix} 8 & 1 & 0 \\ 0 & 8 & 1 \end{pmatrix}, \tag{E.3}$$

and its kernel is again $\langle [1 : 3 : 9] \rangle$.

Proof mode. The invariant/coinvariant construction and the implications of the displayed ranks are conceptual. The dimensions, norm and projection ranks, kernel line, rank-one pencil member, contraction matrix, and flattening census are exact finite linear algebra in the `relative-cubic-depth` bundle and its independent replay.

Proof. Exact invariant and commutator calculations give $\dim R = \dim M_G = 3$, the two ranks in (E.1), and $N\pi = 0$. Thus $\mathrm{im}\, \pi \subseteq \ker N$; both spaces have dimension two, so they are equal. The computed image of N is the displayed kernel of π, proving (E.2).

The invariant symmetric-form pencil on W has a unique rank-one member ℓ^2, where ℓ transforms by χ. Contracting a χ-twisted cubic by ℓ cancels the two characters and lands in the invariant quadratic pencil. Exact contraction gives (E.3), whose kernel proves the second assertion. □

The kernel line is intrinsic rather than an artifact of the displayed basis. Among the 133 projective lines of R, it is the unique line whose first flattening has rank 9; the remaining census is one line of rank 1 and 131 lines of rank 10.

Remark E.2 (Why the two canonical planes do not glue). The three labelled cubic orbit classes project to M_G with relation $[2, 9, 1]$, whereas the three labelled depth profiles have relation $[2, 8, 1]$. There is therefore no label-preserving linear map carrying these source classes to the depth rays. The integral transfer makes the obstruction structural. In orbit-size order set

$$s = (1, 1, 1)^t, \qquad B = s(1, 4, 6).$$

Then $B^2 = 11B$. Divided transfer kills every integral balanced relation a with $(1, 4, 6)a = 0$, but $Bs = 11s$, so it fixes the depth socle after division by 11. It separates the source relation from the depth socle instead of identifying them. The doubled-line and residual-line flag of Remark D.3 is internal to the depth plane and supplies no missing map.

F Verification architecture

Table 1 separates the mathematical arguments from their exact finite inputs. Here "certificate" means a deterministic calculation over a prime field, checked against a canonical JSON output and a SHA-256 manifest; an independent replay uses a separate implementation. "Lean" means that the stated implication is checked by the Lean kernel. Neither term replaces the cited identification of the classical groups or configurations.

Result	Proof in the text or classical input	Exact evidence
Matching-secant quotient	unique conic divisibility and the four-endpoint switch	none
Balanced-orbit completeness	projective–trade bridge, Dickson–Giudici subgroups, Frobenius-digit socles, and the characteristic-three axis trade	generic local first-wall checker and independent closed-form replay; exhaustive $q = 9$ check and independent replay
Ranks and the middle Fischer layer	affine cocycle, irreducible Fischer modules; Edge–Dye configurations	quotient matrices as cross-checks; all edge-selected alternating cycles and independent Dickson replay
Balanced sheets and cubic orientation	Propositions 4.1 and 4.2; Lemma 4.3; maximal-isotropic quotient duality	finite moments and ranks as cross-checks; kernel-checked hyperplane-square implication; Gorenstein replay
Six-profile reconstruction	subgroup marks, orbit–stabilizer, and the three-ray lemmas	profile rows and replay; decorated-parent bijection
Modular depth and the Tate plane	projectivity, fixed-point exactness, and standard modular-module input	invariant, coinvariant, and depth calculations with replay
Arithmetic splitting and gluing	displayed root calculations; cited subgroup identifications	Lean gate and normalized certificate/replay

Table 1: Proof modes for the principal results.

The claim-by-claim map is `verification/trust_manifest.json`; the exact statement identity is `verification/statement_identity.json`. From the root of the archival source, the aggregate entry point is

```
python3 verification/verify_release.py
```

It checks the statement identity and all admitted checksum manifests, runs the admitted primary checks and independent replays, elaborates the arithmetic-gluing, Hilbert-symmetry, and hyperplane-square Lean gates, and rebuilds this manuscript through the repository Makefile.

The enumerated review closure accompanying this candidate is pinned by

`verification/evidence_fingerprint.json`.

Its SHA-256 digest is

`1009fe58e6deff0c614525e751f6fc765c1942`
`41061b13b15c40039ffac9b730`.

That fingerprint fixes a normalized manuscript, statement contents, extractor, paper Makefile, verification documentation, the admitted evidence manifests, command vectors, Lean gates, runner, trust manifest, and environment: Python 3.13.12, Lean 4.32.0-rc1, and Mathlib revision

`571b8a8e54219b4d393f75f4b8653fac08197fcc`.

A successful replay ends with

`clebsch factorization release: CHECK OK`.

The normalization replaces only the displayed fingerprint digest and the statement identity's derived full-source hash, avoiding a circular self-hash. This is an explicit review-source allowlist, not a claim to hash every file in the repository or every transitive tool dependency. An immutable public archive locator must be added before release.

The largest formal component is the gate

`RelativeConicArcs.Gates.`
`ClebschArithmeticGluing`.

Its twenty-three terminals check the displayed roots, matching involutions, projective orders, determinant halves, the A_3/B_3 orbit calculations, and the split–fused conclusion. For H_3, Lean checks the normalized certificate rows, intersections, determinant classes, the $11 + 11$ signature split, and the shape of the bounded generation words. Evaluation and completeness of the 660-element coset and word certificates remain exact generator/replay obligations. The classical names S_4, A_4, A_5, and $\mathrm{PSL}_2(11)$ are cited inputs: no group name is inferred from an order.

The separate structural gate

`RelativeConicArcs.Gates.`
`ClebschHilbertSymmetry`

retains an independent check of the v1 Hilbert arithmetic: a symmetric finite Hilbert function of total length $2q$ whose first values are $1, q-1, q-1$ has socle degree three and final value one. The v2 Gorenstein proof does not use this implication; it obtains the perfect degree-1-by-degree-2 pairing directly from maximal isotropy.

The other computations are load-bearing only at the points shown in Table 1. The matching-module bundle supplies the three finite quotient ranks and cross-checks the moment data. The H_3-equivariant-rank bundle checks the Sylow-normalizer weights, A_5-fixed dimensions, sheet membership, and the top witness. The generic first-wall bundle constructs only S, T, R, Y, verifies the normalized trace, unique spill, Borel gap, and target dimensions, and replays the formulas independently; the Lucas-socle and adjacent-wall identifications remain the human theorem. The small-field trade bundle exhausts $q = 9$; its larger-field rows are cross-checks only. The shared-radial bundle reconstructs every edge-selected pair, its alternating cycle, and its deepest trace; an independent replay starts from the Dickson recurrence. The balanced-sheet bundle independently reconstructs the permutation-module conclusions, while the Gorenstein bundle checks the ideals, deletion ranks, socles, and resolutions. The profile-incidence bundle computes one row per double coset; the decorated-parent bundle identifies a singleton matching with its parent; the relative-cubic-depth bundle computes the modular and Tate rank patterns; and the arithmetic-gluing gate checks the normalized splitting data. All use exact prime-field arithmetic. The balanced-orbit classification uses no field census: its executable component checks the local formulas after the human modular identifications. The general divisibility theorem, radical–Hadamard and hyperplane-square lemmas, modular-permutation mechanism, three-ray cubic lemma, and mass-zero factorization are proved in the text.

This verification surface proves only the three marked configurations defined here. It does not supply an all-characteristic construction, a canonical map from the relative-cubic Tate plane to the depth plane, or any passage, holonomy, theta, Fourier, quantum, or Mathieu identification.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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